

• Original Paper •

Global Ensemble Weather Prediction from a Deep Learning–Based Model (Pangu-Weather) with the Initial Condition Perturbations of CMA-GEPS

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ABSTRACT

Pangu-Weather (PGW), trained with deep learning–based methods (DL-based model), shows significant potential for global medium-range weather forecasting. However, the interpretability and trustworthiness of global medium-range DL-based models raise many concerns. This study uses the singular vector (SV) initial condition (IC) perturbations of the China Meteorological Administration’s Global Ensemble Prediction System (CMA-GEPS) as inputs of PGW for global ensemble prediction (PGW-GEPS) to investigate the ensemble forecast sensitivity of DL-based models to the IC errors. Meanwhile, the CMA-GEPS forecasts serve as benchmarks for comparison and verification. The spatial structures and prediction performance of PGW-GEPS are discussed and compared to CMA-GEPS based on seasonal ensemble experiments. The results show that the ensemble mean and dispersion of PGW-GEPS are similar to those of CMA-GEPS in the medium range but with smoother forecasts. Meanwhile, PGW-GEPS is sensitive to the SV IC perturbations. Specifically, PGW-GEPS can generate realistic ensemble spread beyond the sub-synoptic scale (wavenumbers ≤ 64) with SV IC perturbations. However, PGW’s kinetic energy is significantly reduced at the sub-synoptic scale, leading to error growth behavior inconsistent with CMA-GEPS at that scale. Thus, this behavior indicates that the effective resolution of PGW-GEPS is beyond the sub-synoptic scale and is limited to predicting mesoscale atmospheric motions. In terms of the global medium-range ensemble prediction performance, the probability prediction skill of PGW-GEPS is comparable to CMA-GEPS in the extratropic when they use the same IC perturbations. That means that PGW has a general ability to provide skillful global medium-range forecasts with different ICs from numerical weather prediction.

Key words: deep learning, ensemble prediction, forecast uncertainty, initial condition perturbations, CMA-GEPS, Pangu-Weather

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Article Highlights:

- The DL-based model PGW is sensitive to the SV IC perturbations which can generate realistic ensemble spread beyond the sub-synoptic scale.
- The effective resolution of PGW is beyond the sub-synoptic scale, limited to predicting mesoscale atmospheric motions.
- The global probability prediction skill of PGW-GEPS is comparable to the operational GEPS with the same IC perturbations.

1. Introduction

Global medium-range weather prediction (forecasts valid 3–15 days ahead) is essential for weather forecasting ser-

vices. This is because many high-impact weather and compound extreme events, such as extreme heat, drought, and tropical cyclones, are associated with atmospheric circulation at medium-range time scales (Zscheischler and Seneviratne, 2017). Numerical weather prediction (NWP) is a quantitative and deductive approach that can predict global medium-range weather based on mathematical equations and physical

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laws (Bauer et al., 2015; Shen et al., 2020). With the “quiet revolution” of NWP, the global ensemble prediction system (GEPS) becomes an important component in many NWP centers, as it can estimate the forecast uncertainty caused by initial and model errors based on the temporal evolution of forecast state probability density functions (Leutbecher and Palmer, 2008; Buizza, 2019). The emergence of GEPSs has not only helped forecasters and governments make better decisions, but also enhanced understanding of atmospheric predictability (Zhu et al., 2002; Bauer et al., 2015; Palmer, 2019; Zhang et al., 2019a; Žagar and Szunyogh, 2020).

In recent years, the rapid progress of deep learning-based (DL-based) models has shown massive potential for global medium-range weather forecasting (Bi et al., 2023; Lam et al., 2023; Ben Bouallègue et al., 2024). Unlike traditional NWP models, which are disciplined by physics-based equations, DL-based forecasting models use neural networks or other DL methods to learn weather forecasts directly from training data (usually using reanalysis and analysis datasets). Thus, this inductive approach is also called “data-driven” modeling (Weyn et al., 2019; Rasp et al., 2020). For example, Bi et al. (2023) used Three-Dimensional Earth-Specific Transformers (3DEST) to train a deterministic global forecast model from the ERA5 reanalysis (Hersbach et al., 2020), called Pangu-Weather (PGW), with a horizontal resolution of 0.25° and 13 pressure levels. In terms of its global medium-range forecast performance, the deterministic verification scores of PGW are comparable to those of the Integrated Forecasting System of the European Centre for Medium-Range Weather Forecasts (ECMWF); and notably, PGW can generate a 24-hour global weather forecast in 1.4 seconds. Besides PGW, other DL-based models also show good performance in global medium-range weather forecasting from verification scores (Bi et al., 2023; Chen et al., 2023; ; Hu et al., 2023; Lam et al., 2023). Thus, in conclusion, the significant advantages of DL-based models are that they provide comparable global medium-range weather forecasts with faster inference speed compared to traditional NWP models (Ben Bouallègue et al., 2024). Both these advantages make DL-based models an attractive prospect for extending ensemble members and providing additional information about global medium-range forecast uncertainty.

However, the interpretability and trustworthiness of DL-based models raise many concerns and challenges (Bauer et al., 2023; Huang et al., 2024). For instance, Bonavita (2024) found that PGW, and possibly similar DL-based models, do not have the fidelity and physical consistency of physics-based models, which leads to dynamical balance relationships and other quantities that can be diagnosed or inferred from the mass field and horizontal flow to be unrealistic. Selz and Craig (2023) designed small-amplitude initial condition (IC) perturbations generated from ensemble data assimilation (Isaksen et al., 2010) and documented that PGW cannot produce realistic error growth characteristics from small-amplitude IC perturbations during short-range forecast lead times (0–3 days) (i.e., PGW cannot simulate

the butterfly effect). These findings point to some limitations of DL-based models and provide a scientific basis for further improvement. In summary, the interpretability and trustworthiness of global medium-range DL-based models need further research, and not only from deterministic verification metrics.

Meanwhile, the IC error growth properties and impacts on the forecast sensitivity of DL-based models are still unclear during medium-range forecast lead times. According to current knowledge on the forecast uncertainty in NWP, initial errors can grow easily as baroclinic systems develop, thus affecting the global medium-range flow (Toth and Kalnay, 1993; Buizza and Palmer, 1995). The singular vector (SV) is one of the feasible IC perturbation methods to quantify initial uncertainty that is primarily responsible for forecast error growth associated with baroclinic instability (Molteni et al., 1996; Leutbecher and Palmer, 2008). For the time being, SVs are widely used in generating initial perturbations (IPs) for GEPSs, like those of the ECMWF (Leutbecher and Palmer, 2008), Japan Meteorological Agency (Yamaguchi et al., 2018), and China Meteorological Administration (CMA) (Chen and Li, 2020). This is because SVs can capture dynamically unstable perturbations with realistic growth characteristics and ensure global medium-range ensemble prediction skills (Buizza et al., 2008; Palmer, 2019). Consequently, SV IC perturbations could serve as an efficient tool in investigating the IC error growth and forecast sensitivity of DL-based models during medium-range forecasts.

In this paper, we use a DL-based model, PGW, for global ensemble medium-range weather prediction (PGW-GEPS) with SV IC perturbations, and the results of CMA-GEPS serve as a benchmark to examine several questions: (1) Are the global medium-range forecasts of DL-based models sensitive to IC errors? (2) What is the difference between PGW-GEPS and operational GEPS forecasts when using the same ICs and perturbations? The results could be beneficial in enhancing the interpretability and trustworthiness of global medium-range DL-based models.

Following this introduction, section 2 describes the configurations of PGW and CMA-GEPS. Section 3 introduces the SV IC perturbation method, analysis methods, verification metrics, and design of the global medium-range ensemble experiments. Results are presented in section 4. Finally, section 5 provides a summary and some further discussion. The acronyms related to the NWP system at the CMA and the perturbation procedures used in this study are listed in Table 1.

2. Model description

2.1. Brief introduction to Pangu-Weather

PGW is a deterministic DL-based weather forecast model that uses transformers to learn global weather forecasts from 1979–2017 ERA5 reanalysis datasets [see Bi et al. (2023) for details of PGW; see Hersbach et al. (2020) for details of ERA5], the main configurations of Pangu-Weather are shown in Table 2. According to this strategy,

Table 1. Acronyms for the key terms and procedures used in this study.

(a) Related to the NWP system at the CMA	
GFS	Global Forecast System
4DVAR	Four-Dimensional Variational Data Assimilation System
GEPS	Global Ensemble Prediction System
(b) Related to the perturbation method	
IC perturbation	Initial condition perturbation
TLM	Tangent linear model
ADM	Adjoint model
SV	Singular vector
LPO	Local projection operator
TE	Total energy
OTI	Optimal time interval

Table 2. Main configurations of Pangu-Weather.

Pangu-Weather	
Forecast area	Global
Horizontal resolution	0.25°
Pressure levels	1000 hPa, 925 hPa, 850 hPa, 700 hPa, 600 hPa, 500 hPa, 400 hPa, 300 hPa, 250 hPa, 200 hPa, 150 hPa, 100 hPa and 50 hPa (for upper-air variables)
Input and output variables	u component of wind (u), v component of wind (v), temperature, geopotential (z) and specific moisture (upper-air variables)
Deep networks	10-m u wind (10-m u), 10-m v wind (10-m v), 2-m temperature and mean sea level pressure (surface variables)
	1 h, 3 h, 6 h or 24 h

the number of iterations in the DL model can significantly decrease, which can minimize the root-mean-square error (RMSE) during the forecast lead time to make better forecasts. In this study, we use the 24-h deep network of PGW for global medium-range weather forecasting. Meanwhile, we do not use any transfer learning or fine-tuning to change any parameters in PGW. This is because without using training data as ICs, the generalization ability of PGW can be recognized.

2.2. Brief introduction to CMA-GEPS

CMA-GEPS version 1.3 was independently developed by the CMA's Center for Earth System Modeling and Prediction (CEMC) (See Table 3 for details), based on the CMA Global Forecast System (CMA-GFS version 3.3), and has been in operation since 2018 (Li and Liu, 2019; Chen and Li, 2020; Shen et al., 2020). CMA-GEPS can produce global medium-range ensemble forecasts to estimate the forecast uncertainty during the forecast lead time. In this study, CMA-GEPS serves as a benchmark to verify the trustworthiness of the results generated by PGW.

Table 3. Main configurations of CMA-GEPS version 1.3 in the operation.

CMA-GEPS	
Forecast region	Global
Resolution	0.5°, 87 model vertical levels (model top at 0.1 hPa)
Initial condition	CMA-4DVAR analysis (Zhang et al., 2019a) (dynamical upscaling to 0.5°) (Huo et al., 2018)
Initial perturbation	Singular vector (Liu et al., 2013, 2018)
Model perturbation	SPPT (Peng et al., 2022) and SKEB (Peng et al., 2019)
Ensemble sizes	1 control + 30 perturbed members
Initialized time	0000 UTC and 1200 UTC
Forecast length	15 days
Output frequency	3 h (0–84 h); 6 h (84–360 h)

3. Methodology

3.1. Singular vector initial condition perturbation method

This section briefly introduces the definition and computation algorithm for IC perturbation based on SVs in CMA-GEPS. Due to the fact that small analysis errors can grow rapidly as baroclinic systems develop, which leads to significant forecast uncertainty during the medium-range forecast lead time, to estimate the initial uncertainty, the SV method is designed to capture fast-growing perturbations associated with baroclinic instability in the ICs that are largely responsible for the forecast uncertainty. The computation of SVs can be defined as a maximization problem over an optimal time interval (OTI). In practice, the solutions of SVs are considered as an eigenvalue problem:

$$(\mathbf{E}^{-1}\mathbf{L}^T\mathbf{P}^T\mathbf{E}^2\mathbf{P}\mathbf{L}\mathbf{E}^{-1})\hat{\mathbf{X}} = \lambda\hat{\mathbf{X}}, \quad (1)$$

where λ is the eigenvalue of the matrix $(\mathbf{E}\mathbf{M}\mathbf{P}\mathbf{E}^{-1})^T(\mathbf{E}\mathbf{M}\mathbf{P}\mathbf{E}^{-1})$; $\mathbf{M}$ is the CMA-tangent linear model (TLM); $\mathbf{M}^T$ is the adjoint model [see Liu et al. (2019) for details of CMA-TLM]; $\mathbf{P}$ is the local projection operator, making it possible to compute SVs in the target areas; and $\mathbf{E}$ is a diagonal matrix that defines weighting factors based on the dry-total energy (TE) norm [see Liu et al. (2013) for details of the dry-TE norm in CMA-GEPS] to transform perturbation vector $\mathbf{X}$ into a dimensionless vector $\hat{\mathbf{X}}$ in Eulerian space. Finally, the matrix $\mathbf{E}\mathbf{M}\mathbf{P}\mathbf{E}^{-1}$ is the target SV with the fastest growth rate.

Subsequently, it is essential to combine all SVs to generate optimal IC perturbations that reflect initial uncertainties as much as possible with finite ensemble members. This process starts by rescaling each set of SVs $\hat{\mathbf{X}}$ to $\mathbf{X}$ according to the amplitude of the analysis error:

$$\mathbf{X} = \hat{\mathbf{X}}\left(\frac{\gamma}{\bar{L}}\right), \quad (2)$$

where γ is an empirical parameter to determine perturbation amplitudes and $\bar{L}$ represents the average estimated analysis error from CMA-4DVAR.

After the rescaling procedure, the IC perturbations $\mathbf{M}_j$ [Eq. (3)] consist of a linear combination of $\mathbf{X}_{\text{INI}}$ (initial SV) at the initial time and evolved SVs $\mathbf{X}_{\text{EVO}}$ (evolved SV) at the evolved time:

$$\mathbf{M}_j = \sum_{k=1}^Z \alpha_{j,k} [\beta \mathbf{X}_{\text{INI}} + (1 - \beta) \mathbf{X}_{\text{EVO}}], \quad (3)$$

where j and k denote the number of IC perturbations and SVs, respectively; Z denotes the number of $\mathbf{X}_{\text{INI}}$ and $\mathbf{X}_{\text{EVO}}$; $\alpha_{j,k}$ denotes the random coefficients according to the Gaussian distribution; and the weight coefficients β determine the amplitude ratio between the initial and evolved SVs. The initial SVs deliver rearward sloping structures associated with baroclinic instability. This characteristic makes small IC perturbations grow rapidly and realistically to reach synoptic scales responsible for a large part of the forecast uncertainty (Leutbecher and Palmer, 2008; Li and Liu, 2019; Liu et al., 2024). The evolved SVs represent perturbations that have been growing during the current and past data - assimilation cycles (48 h prior to the initial SV), which are related to the analysis uncertainty (Buizza et al., 2008; Li and Liu, 2019). Mostly, β (0.9) is relatively larger to ensure that the amplitude of initial SVs is dominant (Wang et al., 2020). The calculation settings for SV IC perturbations are summarized in Table 4. Therefore, $\mathbf{P}_j$ is the optimal IC perturbation we need. In this paper, we use 31 IC perturbations (one unperturbed control member + 30 members) consisting of a pair of 15 SV IC perturbations (plus and minus with CTL).

In this paper, we use the 2.5° grid-spacing of CMA-TLM (ICs from CMA-4DVAR dynamical upscaling) with 48 h OTI to calculate SVs representing dynamically unstable IC perturbations that are responsible for the largest forecast uncertainty (see Table 4 for details). The main reasons for using these SV IC perturbations are as follows:

(1) DL-based global weather forecasting models (i.e., PGW) could provide skillful forecasts on synoptic scales in the medium range (Bonavita, 2024; Selz and Criag, 2023; Ben Bouallègue et al., 2024).

(2) The SV IC perturbations can grow rapidly to

develop synoptic-scale structures associated with baroclinic instability, ensuring the spread of outcomes grows realistically in the medium range (Leutbecher and Palmer, 2008; Chen and Li, 2020).

(3) Compared to ensemble data assimilation perturbations, SV perturbations have relatively larger scales at the initial time that PGW could resolve (Buizza et al., 2008), which provides the possibility for DL-based models to capture the temporal evolution of SV IC perturbations. Consequently, SV IC perturbations could be property candidates for investigating the forecast sensitivity of DL-based models.

It should be noted that the only difference in this paper between the IC perturbations for CMA-GEPS and PGW-GEPS is their horizontal resolution. For CMA-GEPS, 2.5° SVs are interpolated to 0.5° , with the same resolution as the CTL (CMA-4DVAR dynamical upscaling) in CMA-GEPS. However, PGW requires 0.25° input data for inferencing. Therefore, for PGW-GEPS, 2.5° SVs are interpolated to 0.25° , and then initialized by CMA-GEPS to convert IC perturbations from model levels to pressure levels. Moreover, both CMA-GEPS and PGW-GEPS could resolve 2.5° SVs because the horizontal resolution of SVs is larger than $3\Delta x$ (horizontal spacing Δx) compared to CMA-GEPS and PGW-GEPS.

3.2. Design of the ensemble sensitivity experiments

Following the construction of SV IC perturbations (section 3.1), we conducted two ensemble experiments, CMA-GEPS and PGW-GEPS, based on the SV IC perturbations (see details in Table 5). The ensemble forecasts of CMA-GEPS were used as the benchmark for comparing the trustworthiness and prediction performance of PGW-GEPS. This is because all the results of CMA-GEPS were generated operationally, and the probabilistic prediction results of CMA-GEPS are evaluated based on the World Meteorological Organization (WMO) guidelines for ensemble prediction systems (Mylne et al., 2022), which are available from the Lead Centre on Verification of Ensemble Prediction Systems website. Notably, the IC perturbations of PGW-GEPS only include SVs (from CMA-GEPS) and lack a model perturbation module. These settings could help us investigate the ensemble forecast sensitivity of the DL-based model to the

Table 4. Configurations for computing the SVs.

Variables	Description
Target areas	30°–80°N (NH) 30°–80°S (SH)
Resolution	$2.5^\circ/87$ model levels
Perturbed variables in CMA-TLM	Perturbations of CMA-TLM, zonal and meridional wind, perturbed potential temperature, perturbed Exner pressure [see Li and Liu (2019) for details]
OTI	48 h
Norm	Dry total energy norm (TE)
Vertical integration	4th to 50th model level (nearly 100 m to 16 000 m)
Dry linearized physical processes	Subgrid-scale orographic effect, vertical diffusion
No. of SVs	15

Table 5. Main configurations of CMA-GEPS, PGW-GEPS, and PGW-RP-GEPS for ensemble forecast sensitivity analysis.

	CMA-GEPS	PGW-GEPS	PGW-RP-GEPS
Forecast region	Global	Global	Global
Horizontal Resolution	0.5°	0.25°	0.25°
Initial condition	CMA-4DVAR analysis (dynamical upscaling to 0.5°)	CMA-4DVAR analysis	CMA-4DVAR analysis
Initial perturbation	Singular vector (2.5°, 48 h OTI)	Singular vector (2.5°, 48 h OTI)	Random perturbations (RP)
Model perturbation	SPPT and SKEB	/	/
Ensemble sizes	1 control (unperturbed) + 30 perturbed members (plus and minus 15 pair of SVs)	1 control (unperturbed) + 30 perturbed members (plus and minus 15 pair of SVs)	1 control (unperturbed) + 30 perturbed members (plus and minus 15 pair of RPs)
Initialized time	0000 UTC	0000 UTC	0000 UTC
Forecast length	15 days	15 days	15 days
Output frequency	24 h	24 h	24 h

IC errors. Meanwhile, the influence of model perturbations in CMA-GEPS will be discussed later.

Additionally, we designed a set of ensemble forecasts based on random perturbations, referred to as PGW-RP-GEPS, to serve as a comparison in the evaluation of global ensemble prediction performance. The random perturbations' specific configuration and accompanying pseudocode can be found in [Bi et al. \(2023\)](#). To ensure better comparability of the PGW-GEPS, the perturbed variables include geopotential height, temperature, and wind components.

The ensemble forecast sensitivity and performance of CMA-GEPS, PGW-GEPS, and PGW-RD-GEPS are analyzed based on 40 cases. In order to reduce the impact of test cases with sufficient samples ([Zhu et al., 2023](#)), representative months [Jan (202301), Apr (202304), Jul (202307), and Oct (202210)] from different seasons were selected for ensemble experimentation and verification. For each experiment, the ensemble forecasts are initialized every three days (days 1, 4, 7, 10, 13, 16, 19, 22, 25, and 28 for the four months starting at 0000 UTC). The ensemble forecast time is 15 days, and the forecast interval is 24 h.

3.3. Analysis methods and verification metrics

In this study, the ensemble forecast sensitivity and performances of CMA-GEPS and PGW-GEPS are evaluated using a comprehensive set of forecast verification metrics.

First, ensemble forecasting should provide a flow-dependent prediction of the probability distribution of forecast uncertainty. Furthermore, assuming that the forecast uncertainty distribution is Gaussian, the expectation (ensemble mean) and standard deviation (ensemble spread) provide a complete description of the predicted probability distribution. Consequently, following the WMO guidelines ([Mylne et al., 2022](#)), the ensemble mean $\tilde{F}_t$ at forecast lead time t can be defined by Eq. (4) from all the ensemble members $F_{t,j}$:

$$\tilde{F}_t = \frac{1}{N} \sum_{j=1}^N F_{t,j}, \quad (4)$$

where N represents the total size of the ensemble members,

including both the CTL and perturbed members. This implies that the $\tilde{F}_t$ can be used to represent the prediction of the mean of the probability distribution of the forecast uncertainty at the lead time. Subsequently, the ensemble spread S_t can be defined as

$$S_t = \sqrt{\frac{1}{N} \sum_{j=1}^N (F_{t,j} - \tilde{F}_t)^2}. \quad (5)$$

S_t describes the degree of deviation of the ensemble members from the ensemble mean at the lead time. Furthermore, S_t can also be regarded as the forecast uncertainty ([Žagar, 2017](#)). Correspondingly, the forecast error (RMSE) at lead time E_t can be considered a diagnostic tool for measuring the difference between a true state A_t (analysis or reanalysis) and the ensemble mean of the probability distribution $\tilde{F}_t$:

$$E_t = \sqrt{(\tilde{F}_t - A_t)^2}, \quad (6)$$

where E_t is typically regarded as the magnitude of the forecast uncertainty ([Zhu et al., 2002](#)). Besides, E_t and S_t should ideally satisfy the approximate equality.

Based on the definitions of the ensemble mean and ensemble spread, the forecast sensitivity of models can be estimated by the difference kinetic energy (DKE; [Zhang et al., 2003](#)):

$$\text{DKE}_{p,t} = \frac{1}{2} (\Delta u_{p,t}^2 + \Delta v_{p,t}^2), \quad (7)$$

where p denotes the pressure level. Note that in previous studies on atmospheric predictability, Δ usually indicates the difference between the CTL and perturbed member ([Judt, 2020](#); [Rotunno et al., 2023](#)). However, we want to investigate the lead time of the forecast sensitivity with respect to the mean square of the ensemble spread (SDKE) and ensemble mean forecast error (EDKE) based on the DKE. Thus, the SDKE can be defined based on Eq. (5) and Eq. (7):

$$\text{SDKE}_{p,t} = \frac{1}{2N} \sum_{j=1}^N [(F_{u,p,t,j} - \widetilde{F}_{u,p,t})^2 + (F_{v,p,t,j} - \widetilde{F}_{v,p,t})^2], \quad (8)$$

and the EDKE can be defined based on Eq. (6) and Eq. (7):

$$\text{EDKE}_{p,t} = \frac{1}{2} [(\widetilde{F}_{u,p,t} - A_{u,p,t})^2 + (\widetilde{F}_{v,p,t} - A_{v,p,t})^2]. \quad (9)$$

SDKE and EDKE have been designed to represent the forecast sensitivity of the ensemble spread and ensemble mean forecast error according to the kinetic energy (KE), respectively.

To investigate the forecast sensitivity of spatial scale, the global KE, SDKE, and EDKE are calculated from the global horizontal wind fields using the spherical harmonics function. For clarity, the different spatial scales are defined in Table 6, which has been widely used in KE spectral analyses of global NWP (Wang and Sardeshmukh, 2021; Li et al., 2024b). Subsequently, the KE is decomposed into its rotational and divergent components in the spherical harmonic space to gain insight into the forecast sensitivity regarding the underlying governing dynamics. Meanwhile, the forecast sensitivity of KE spectra with respect to the tropical region is obtained by multiplying the $\cos^{30}(\varphi^2)$ (φ denotes latitude) global fixed grid (0.5°) before spherical harmonic decomposition [$1 - \cos^{30}(\varphi^2)$ for the extratropics] [see Wang and Sardeshmukh (2021) and Li et al. (2024b) for details of the regional division].

Lastly, the global ensemble prediction performances of CMA-GEPS and PGW-GEPS are evaluated following the WMO guidelines (Mylne et al., 2022). For fair verification, both CMA-GEPS and PGW-GEPS forecasts are verified against the CMA-4DVAR operational analysis [see Zhang et al. (2019b) for details of CMA-4DVAR]. Besides, the verification areas are divided into the Northern Hemisphere (NH, 90° – 20° N, inclusive, all longitudes), Southern Hemisphere (SH, 90° – 20° S, inclusive, all longitudes), and tropics (Trpc, 20° N– 20° S, inclusive, all longitudes) (Buizza et al., 2005; Mylne et al., 2022). In addition, all the forecasts are interpolated to 1.5° grid resolution, which scales both CMA-GEPS and PGW-GEPS can resolve following the WMO guidelines (Mylne et al., 2022). All the forecast metrics are calculated from all cases (total 40) for average, and a bootstrap and Student's t -test algorithm (Hamill, 1999; Zhu et al., 2023) are used to test the significance of the score differences between CMA-GEPS and PGW-GEPS.

Table 6. Definition of the spatial scales according to total spherical wavenumbers.

	Total spherical wavenumber
Large scales	≤ 20
Synoptic scales	20–63
Sub-synoptic scales	64–100
Mesoscales	101–160

4. Results

4.1. Uncertainty between the control forecast of CMA-GEPS and ERA5 at the initial time

In this section, we discuss the uncertainty between the CTL of CMA-GEPS and ERA5 at the initial time. This is because PGW learns global weather forecasts based on the long-period historical ERA5 reanalysis datasets. However, CMA-GEPS-CTL is used as the IC for PGW inference in this study. To better understand the forecast sensitivity of PGW-GEPS, it is necessary to recognize the uncertainty between the CTL of CMA-GEPS and ERA5.

Table 7 describes the Pearson correlation coefficient between CMA-GEPS-CTL and ERA5 in different target areas at the initial time. It should be noted that we interpolate the ERA5 grid spacing from 0.25° to 0.5° (the same as that of CMA-GEPS-CTL). Furthermore, the geopotential z of ERA5 is converted to the geopotential height (GH) for comparison with CMA-GEPS. Specifically, CMA-GEPS-CTL is similar to ERA5, especially for variables GH and t , with coefficients larger than 0.99 in the middle (500 hPa) to the high (250 hPa) troposphere. At the same time, most of the wind field coefficients are above 0.9. Inductively, the analysis fields of CMA-GEPS-CTL are comparable to ERA5 in Global, NH, and SH. However, Table 7 also shows that the analysis uncertainty between CMA-GEPS-CTL and ERA5 mainly exists in Trpc, especially for v wind fields. The lack of observations in Trpc leads to the large uncertainty in the data assimilation.

Figure 1 displays the average KE spectra of ERA5 and CMA-GEPS-CTL at the initial time. First, ERA5 is interpolated to a horizontal resolution of 0.5° for comparative analysis with CMA-GEPS-CTL. Additionally, due to the effectiveness of the model resolution, the KE spectra are truncated at wavelengths greater than $3\Delta x$ (total wavenumber less than

Table 7. Pearson correlation coefficient between the analysis of CMA-GEPS-CTL and ERA5 reanalysis at the initial time, calculated from the average of all cases at the initial time with significance testing. Aside from the whole globe (i.e., “Global”), the different regions are: Northern Hemisphere (NH, 90° – 20° N, inclusive, all longitudes); Southern Hemisphere (SH, 90° – 20° S, inclusive, all longitudes); and the tropics (Trpc, 20° N– 20° S, inclusive, all longitudes).

Variable	Pressure levels	Global	NH	SH	Trpc
GH	250 hPa	1.000	1.000	1.000	1.000
	500 hPa	1.000	1.000	1.000	1.000
	850 hPa	0.999	1.000	0.999	1.000
t	250 hPa	1.000	1.000	1.000	1.000
	500 hPa	1.000	1.000	1.000	1.000
	850 hPa	0.988	0.993	0.993	0.963
u	250 hPa	0.988	0.993	0.993	0.963
	500 hPa	0.968	0.981	0.981	0.945
	850 hPa	0.945	0.936	0.963	0.909
v	250 hPa	0.972	0.988	0.987	0.918
	500 hPa	0.929	0.966	0.970	0.787
	850 hPa	0.916	0.928	0.953	0.815

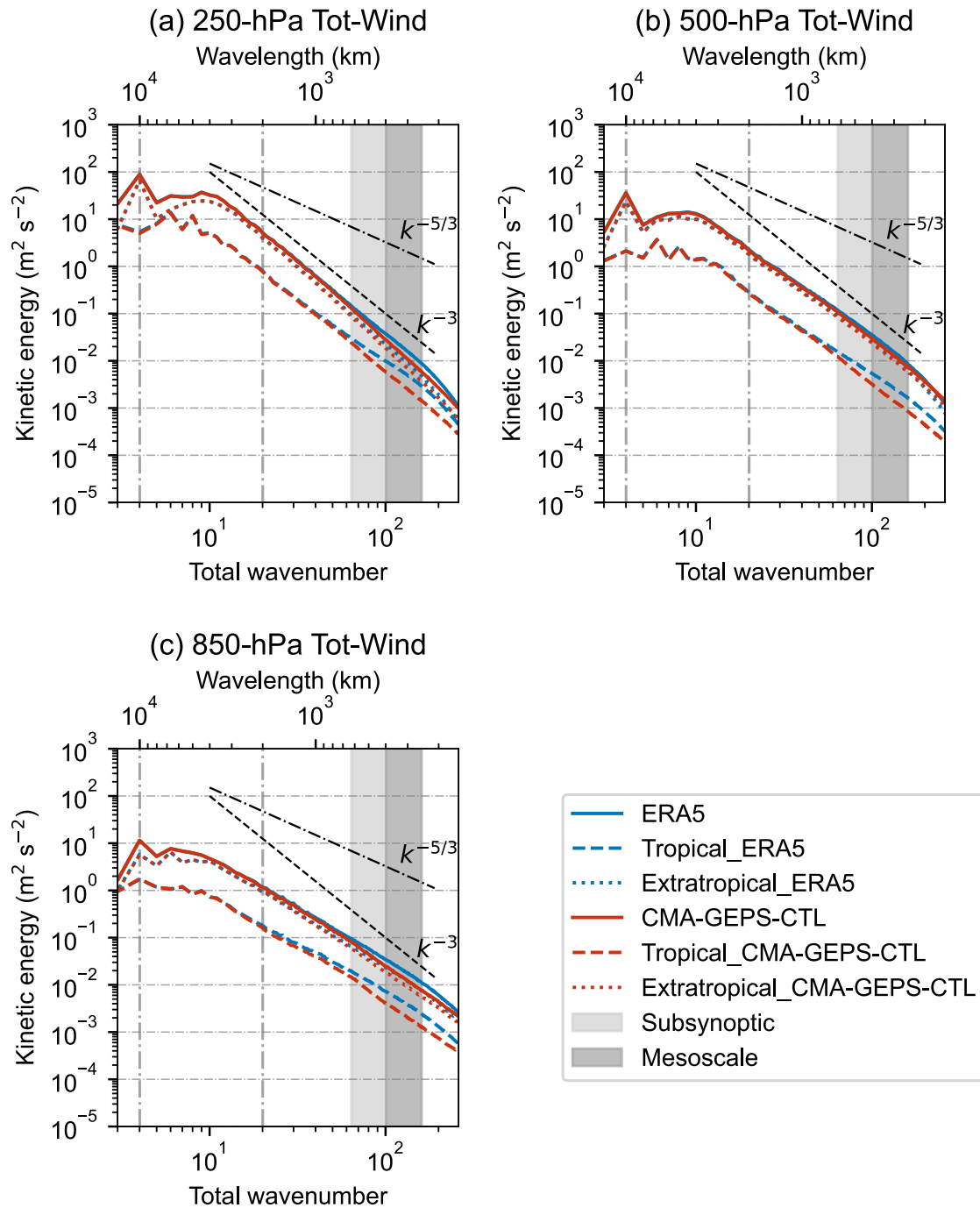


Fig. 1. KE spectra at the initial time of ERA5 (blue) and CMA-GEPS-CTL (red), in which the solid line represents the global region, the dashed line the tropical region, and the dotted line the extratropical region. The spectra are displayed across the isobaric surfaces of (a) 250 hPa, (b) 500 hPa, and (c) 850 hPa. The sub-synoptic-scale [total wave number (n): $64 \leq n \leq 100$] and mesoscale [total wave number: $101 \leq n \leq 160$] ranges are indicated. Slope lines of $-5/3$ and -3 are marked in each subplot for reference, calculated from the average of all cases.

or equal to 260) for analysis. Furthermore, to more clearly illustrate the differences in the KE spectra within the spatial scale, Fig. 1 displays the structural features of the KE spectra for total wavenumbers from 3 to 260 since the large-scale structures of the KE spectra from ERA5 and CMA-GEPS-CTL closely align. Therefore, the subsequent KE spectral analysis will adopt the settings outlined above.

It can be seen from Fig. 1 that the structure of the KE spectra of ERA5 and CMA-GEPS-CTL are highly similar. For example, as the isobaric surface increases, the global KE spectra become closer to -3 , whereas as the isobaric surface height decreases, the global KE spectra approach $-5/3$ to -3 . Furthermore, the amplitude and characteristics of the extratropical KE spectra are close to those of the global KE

spectra. Additionally, the KE spectra of rotational wind (figures not shown for brevity) remain similar to the structures of the total wind, while the slopes of the KE spectra of the divergent wind components are close to -3 for different isobaric surfaces (figures not shown for brevity). These differences in KE spectral slopes are related to the underlying governing dynamics, such that the unbalanced flow fits well with the $-5/3$ slope, especially below the sub-synoptic scale and in the tropics. In contrast, the extratropical atmospheric flow is dominated by balanced motions (e.g., Rossby waves, baroclinic waves) and some localized convection, resulting in slopes ranging from $-5/3$ to -3 . Overall, the structures of the ERA5 and CMA-GEPS-CTL KE spectra are very similar and consistent, with differences in the mesoscale. These differences are mainly due to the divergent wind (figures not shown for brevity).

Furthermore, Fig. S1 [(in the electronic supplementary material (ESM))] illustrates the prediction skill of PGW when used with CMA-GEPS-CTL (shown in red, labeled as PGW-CMA-CTL) and ERA5 (shown in blue, labeled as PGW-ERA5). Fig. S1 (in the ESM) displays the anomaly correlation coefficient (ACC) scores for the 500-hPa geopotential height over NH and SH. The ACC metrics for the 500-hPa geopotential height are widely used to measure global medium-range prediction skills. An ACC value closer to 1 indicates a higher level of prediction skill, while an ACC value falls below 0.6 suggests that the positioning of synoptic scale features no longer provides valuable forecasting. The respective analysis and reanalysis are used as the reference truth to make fair verification.

It can be seen from Fig. S1 (in the ESM) that PGW can provide valuable forecasts for both NH and SH at a lead time of around 7.5 to 8 days when using CMA-GEPS-CTL, compared to 9 days when using PGW-ERA5. This difference is likely because ERA5 can assimilate more observations than CMA-GEPS-CTL, resulting in better analysis quality. Additionally, since PGW was trained on ERA5 datasets generated by the Integrated Forecasting System (IFS) Cy41r2, this ensures better model consistency between the training data and input data, thus leading to higher forecast skills when using ERA5.

However, it should be noted that the ACC score of PGW-CMA-CTL is almost equal to that of CMA-GFS (Shen et al., 2023). This indicates that although PGW is not trained in CMA-GFS analysis, it can effectively capture the evolution of large-scale weather systems, leading to a forecast accuracy comparable to CMA-GFS when using the same analysis as input data. In other words, the prediction skills of PGW and other DL-based models are “model-dependent” when different analyses are used as input data. This finding further emphasizes the impact of uncertainties between training data and real-time data on the predictive ability of DL-based models. Meanwhile, it is necessary to use the same analysis as the input data and reference truth for PGW to make a fair comparison and verification.

In summary, no global NWP model can produce accurate

medium-range forecasts, and the initial uncertainty is unavoidable. Therefore, many NWP centers use ensemble forecasting to estimate the forecast uncertainty in the medium range. Currently, the promising correlation between CMA-GEPS-CTL and ERA5 at the global scale provides the basis for investigating the forecast sensitivity of PGW. In addition, the inevitable uncertainties between the operational analysis and reanalysis are beneficial for understanding PGW’s generalization ability and forecast sensitivity to the initial uncertainty. Furthermore, it is important to use the same analysis as the input data and reference truth for PGW to allow a fair comparison and verification.

4.2. Forecast sensitivity of CMA-GEPS and PGW-GEPS

In this section, we aim to analyze the spatial distribution of the ensemble mean and ensemble spread of PGW-GEPS during the medium-range forecast lead time in an extreme high-temperature event. The results will be compared with those of CMA-GEPS to examine the differences between the two GEPSs, as well as to assess the reasonableness and forecast sensitivity of PGW-GEPS. Subsequently, we will investigate the temporal evolution of the SDKE and SEKE spectra of PGW-GEPS with respect to the forecast lead time from the average of all cases. This analysis will provide insights into the ensemble forecast sensitivity of PGW-GEPS at different spatial scales.

4.2.1. Forecast sensitivity of PGW-GEPS to the IC perturbations

The structures and characteristics of CMA-GEPS and PGW-GEPS are evaluated based on the 15-day ensemble forecasts initialized at 0000 UTC 1 July 2023. During this period, the Beijing Meteorological Service issued red warnings (the highest level) for extremely high temperatures (20230706). This section emphasizes the global medium-range ensemble forecasting of atmospheric circulation based on CMA-GEPS-CTL for verification. Additionally, the national automatic station (Haidian, in Beijing) temperature observation data are used as the truth.

Figure S2 (in the ESM) presents the horizontal and vertical cross-section structures of the SV IPs in the NH at the 500-hPa pressure level of the v wind speed. From Fig. S2a, SV IPs are mainly located in the trough and ridge areas of baroclinic instability at mid-to-high-latitudes at the initial time. In addition, Fig. S2b describes the vertical cross-section structures of the SV IPs, which deliver rearward sloping baroclinic structures at the initial time, and information for the final-time synoptic-scale structure is contained in the initial SVs. Then, SVs with sub-synoptic scale wavenumbers may grow rapidly to reach synoptic scales during the optimal time interval. For clarity, the details of the patterns and temporal evolutions of the calculated SVs in CMA-GEPS can be found in Wang et al. (2020) and Liu et al. (2024). In summary, the SV IPs can reflect the strongest dynamical instability in target areas.

Figure 2 presents the ensemble forecasts of the 500-hPa geopotential height for the 72-h forecasts of CMA-GEPS

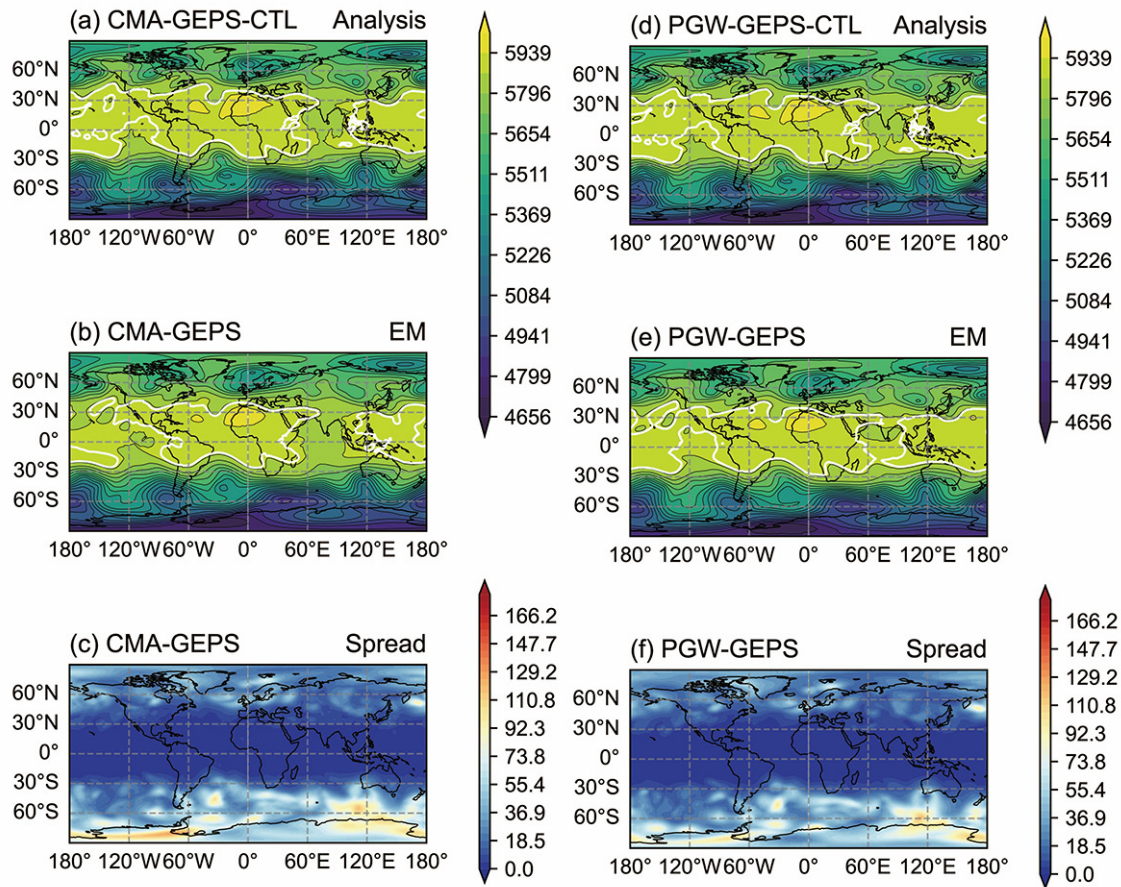


Fig. 2. The 500-hPa geopotential height for the 72-h ensemble forecasts of CMA-GEPS and PGW-GEPS. The first row shows the analysis at the forecast lead time as the reference truth [(a) is the same as (d)]. The second row shows the ensemble mean of (b) CMA-GEPS and (e) PGW-GEPS, with the white line denoting the value of 5880 gpm (subtropical high belts). The third row shows the ensemble spread of (c) CMA-GEPS and (f) PGW-GEPS. The forecasts were initialized at 0000 UTC 1 July 2023.

and PGW-GEPS. The 500-hPa geopotential height can effectively show the primary tropospheric waves that are associated with the dominant weather systems in the medium range. It can be seen that the spatial distribution of the 72-h ensemble mean of CMA-GEPS (Fig. 2b) and PGW-GEPS (Fig. 2e) are very similar to the CMA-4DVAR analysis (Figs. 2a or d). Specifically, both GEPSs can capture the temporal evolution of troughs and ridges in the mid- to high-latitude troposphere in NH and SH. Additionally, CMA-GEPS and PGW-GEPS can reflect the distribution characteristics of subtropical high belts (white lines), which are associated with extreme high-temperature events. However, there is a noticeable difference in that the PGW-GEPS forecast is smoother in the tropics, although its horizontal resolution is higher than that of CMA-GEPS. Thus, PGW-GEPS maintains a general indication of a subtropical high belt but lacks details due to excessive smoothness. For ensemble spread, the spatial characteristics of CMA-GEPS and PGW-GEPS are also similar. Specifically, the ensemble spread of both GEPSs is mainly located in the mid-to-high latitudes in the extratropics, consistent with the locations of cyclones and anticyclones. This is because SV IC perturbations tend to be local-

ized and developed in areas of strong baroclinic instability. However, the ensemble spread of CMA-GEPS exhibits a higher amplitude than PGW-GEPS, likely due to the smoothing behaviors in PGW-GEPS.

Figure S3 (in the ESM) is the same as Fig. 2 but for 240-h ensemble forecasts. Compared to the analysis at the forecast lead time, the ensemble means of both CMA-GEPS and PGW-GEPS exhibit greater differences, indicating decreased forecast skill at this range. Focusing on the spatial distribution of ensemble spread, large values for both CMA-GEPS and PGW-GEPS concentrate over mid-to-high latitudes in SH, and low and mid-to-high latitudes in NH. Since the structure of atmospheric flow tends to be more baroclinically unstable in SH than in NH, this allows the SV IC perturbations to grow more easily and faster over SH. As a result, the regions of large ensemble spread have higher amplitude and extent in SH compared to NH. The similar spatial distribution and amplitude of the spread between CMA-GEPS and PGW-GEPS suggests that the SV IC perturbations in PGW can also reasonably grow to maintain sufficient dispersion. In conclusion, the ensemble forecasts of PGW-GEPS are also sensitive to the dynamically unstable IC perturbations

compared to traditional NWP.

Figure 3 illustrates spaghetti plots for the 500-hPa geopotential height in the 5725- and 5880-gpm ensemble forecasts from CMA-GEPS and PGW-GEPS at 72-, 120-, and 168-h forecast lead times. At the 72-h lead time (Figs. 3a and d), the ensemble forecasts of CMA-GEPS and PGW-GEPS are similar to CMA-GEPS-CTL with small dispersion. It should be noted that CMA-GEPS and PGW-GEPS are sensitive to the SV IC perturbations, especially in the trough and ridge areas. However, the main differences can be observed in the features of the western Pacific subtropical high belts. PGW-GEPS shows a westward shift in the position of the western

Pacific subtropical high due to the bias of PGW-GEPS CTL. At the 120-h lead time (Figs. 3b and e), both GEPSs produce comparable ensemble forecasts for midlatitude atmospheric flow, closely matching the analysis, but PGW-GEPS is smoother. With regard to the western Pacific subtropical highs, the CTL of PGW-GEPS is closer to the analysis with smaller dispersion in the tropics. At the 168-h lead time (Figs. 3c and f), both GEPSs generate ensemble forecasts with more dispersion than before. This indicates that the forecast uncertainty is increasing while the prediction skill is decreasing. Overall, CMA-GEPS and PGW-GEPS both provide comparable probability distributions of forecast uncer-

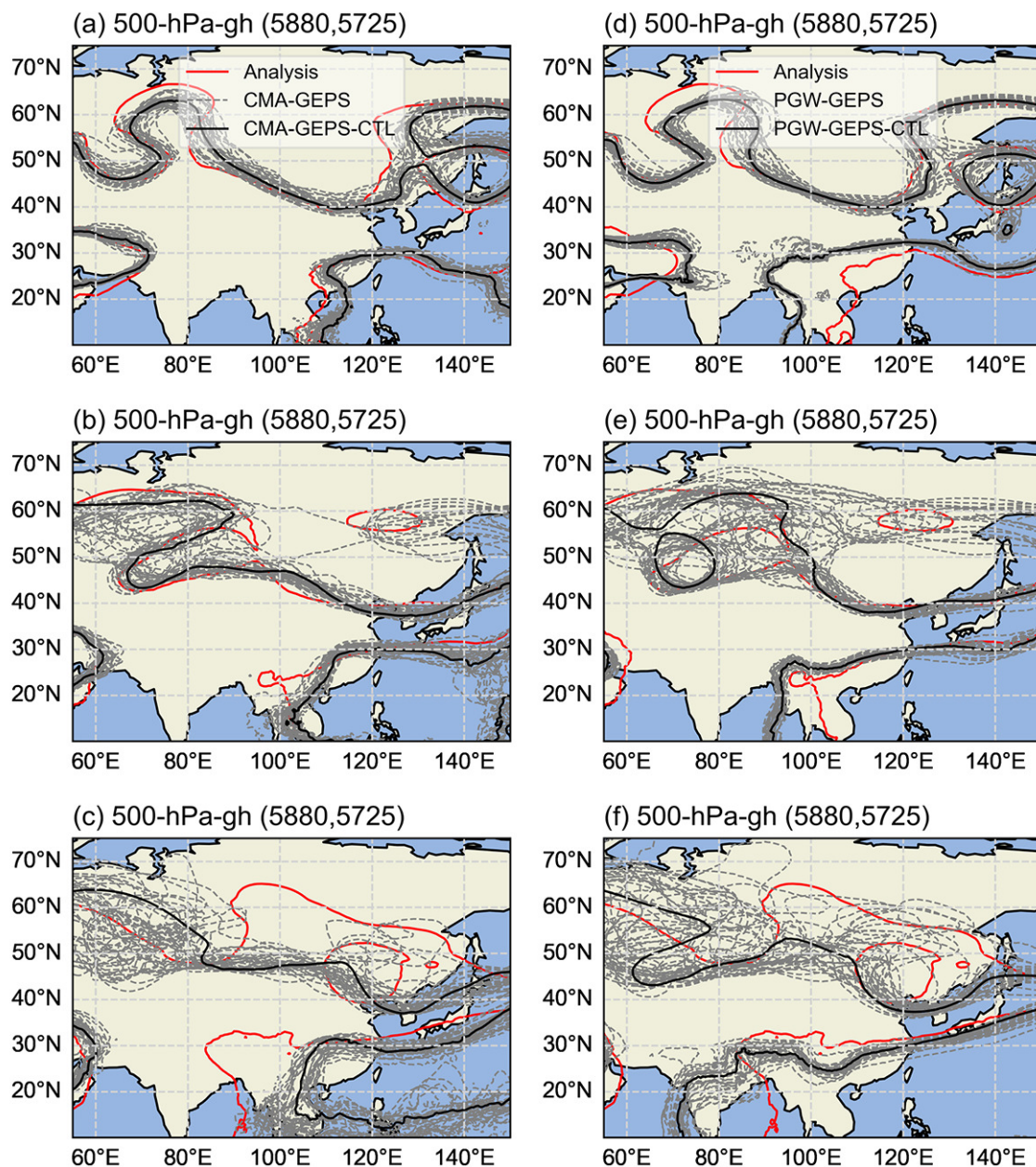


Fig. 3. Spaghetti plots of the 500-hPa geopotential height (5880 and 5725 gpm) for the 72-, 120-, and 168-h ensemble forecasts of (a–c) CMA-GEPS and (d–f) PGW-GEPS. The red line is CMA-GEPS-CTL at the forecast lead time as the reference truth; the black lines are for the control forecast; the gray dashed lines are for the ensemble members. The ensemble forecasts were initialized at 0000 UTC 1 July 2023.

tainty associated with baroclinic instability. The SV IC perturbation can also grow as baroclinic systems develop with sufficient dispersion in PGW-GEPS. In other words, PGW is sensitive to the dynamically unstable IC perturbations during the medium-range forecast lead time.

Figure 4 shows the time series evolution of ensemble forecasts for 2-m temperature from CMA-GEPS and PGW-GEPS compared to observations. From Fig. 4 we can see that CMA-GEPS can show a temperature drop in the short range (forecast lead time less than three days), but all members underestimate the 2-m temperature. CMA-GEPS also shows the temperature rise from a lead time of 3–5 days, but there is still an overestimation. After the fifth day, CMA-GEPS ensemble forecasts diverge more, and the observation is within CMA-GEPS's forecast uncertainty estimate. PGW-GEPS also reflects the temperature drop during the short range, but it is still underestimated compared to CMA-GEPS. Notably, the PGW-GEPS ensemble forecasts are very close to the observation from lead-time day 6 to day 9, and the observation is within PGW-GEPS's forecast uncertainty estimate. After day 10, ensemble forecasts become more dispersed. Nevertheless, in terms of probabilistic forecast skills for 2-m temperature, PGW-GEPS is almost comparable to CMA-GEPS. Meanwhile, the SV IC perturbations are calculated from the upper-air model levels, and the ensemble spread for the 2-m temperature of PGW-GEPS also grows with the increase in lead time. This means that the ensemble forecasts for upper-air and surface variables of PGW are sensitive to dynamically unstable IC perturbations during the medium-range forecast lead time. Meanwhile, variability in ensemble spread in PGW-GEPS shows an adaptive response to forecast uncertainty across different lead times. This is because the 2-m temperature, a diagnostic variable calculated by interpolating between the Earth's surface and bottom model layer, is affected by SV IC perturbations.

4.2.2. Spectral analysis for the forecast sensitivity of PGW-GEPS

In this section, we investigate the forecast sensitivity of PGW-GEPS in spectral space from the average of all cases in order to understand the scale-dependent forecast sensitivity of the DL-based model with the increase in forecast lead time, as well as to uncover the differences between the DL-based model and traditional NWP models.

Figure S4 (in the ESM) displays the temporal evolution of the total wind KE spectra of CMA-GEPS-CTL. Overall, the KE spectra at different altitudes and in different regions remain the main characteristic at the initial time, with the forecast lead time increasing. However, Fig. S4 (in the ESM) shows that the KE spectra exhibit small dissipation below the mesoscale between the forecast lead time and the initial time. The main reason for this scenario can be attributed to the dissipated gravity waves. Concurrently, the KE spectra of rotational and divergent wind exhibit similar structural characteristics to those of total wind (figures not shown for brevity).

Figure S5 (in the ESM) shows the temporal evolution of the total wind KE spectra of PGW-GEPS-CTL. Note that the grid spacing of PGW-GEPS has been interpolated from 0.25° to 0.5° (identical to that of CMA-GEPS). However, there is a discrepancy between the KE spectra of PGW-GEPS-CTL and CMA-GEPS-CTL in the reduction of KE below the sub-synoptic-scale range. Specifically, this reduction is observed in the middle to upper layers (500 and 250 hPa, Figs. S5a and b). Meanwhile, the KE is reducing below the mesoscale range in the lower layers (850 hPa, Fig. S5c). According to those reduction behaviors, all ensemble members of PGW-GEPS are sensitive to the forecast lead time. Furthermore, this reduction is also related to the different regions. As a result of the reduction, the forecast lead evolution of KE spectra of PGW-GEPS-CTL is inconsistent

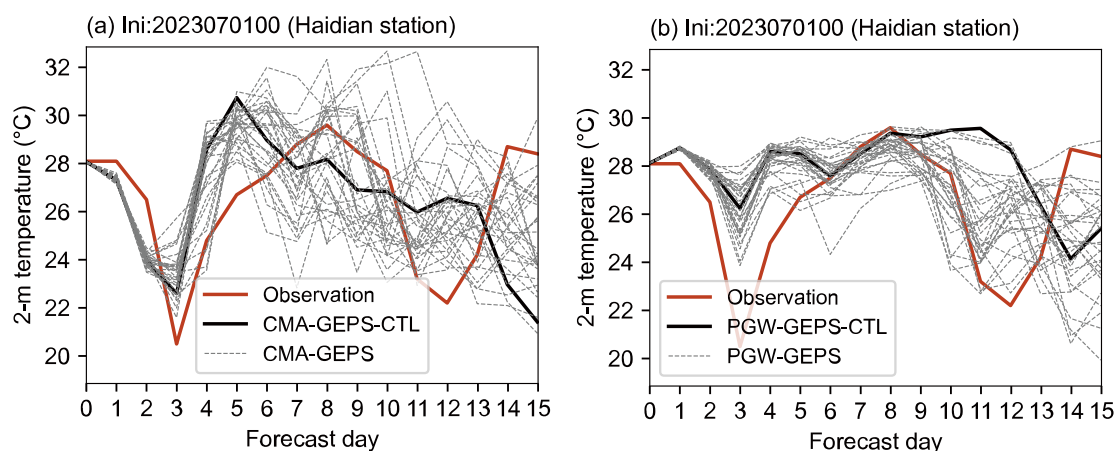


Fig. 4. Time series of the 2-m temperature ensemble forecasts of (a) CMA-GEPS and (b) PGW-GEPS. The red line is for the observation data from the national automatic station (Haidian, in Beijing); the black line is for the control forecast; the gray dashed lines are for the ensemble members. The ensemble forecasts were initialized at 0000 UTC 1 July 2023.

with CMA-GEPS-CTL.

The graph in Fig. S6 (in the ESM) indicates that the KE spectra of CMA-GEPS are similar to CMA-GPES-CTL, with the lead time increasing. Given the characteristics of KE spectra, the initial and model perturbations cannot change the main structures of CMA-GEPS-CTL. Model perturbations barely change the spectra of CMA-GEPS, owing

to the small amplitude of SPPT and SKEB (discussed with the developer of model perturbation in CMA-GEPS). Overall, the KE spectra of CMA-GEPS are consistent with those of CMA-GEPS-CTL, which is without any perturbations.

The KE spectra of PGW-GEPS (Fig. 5) can be compared with CMA-GEPS in Fig. S6 (in the ESM), showing that the KE spectra of PGW-GEPS are also inconsistent with CMA-

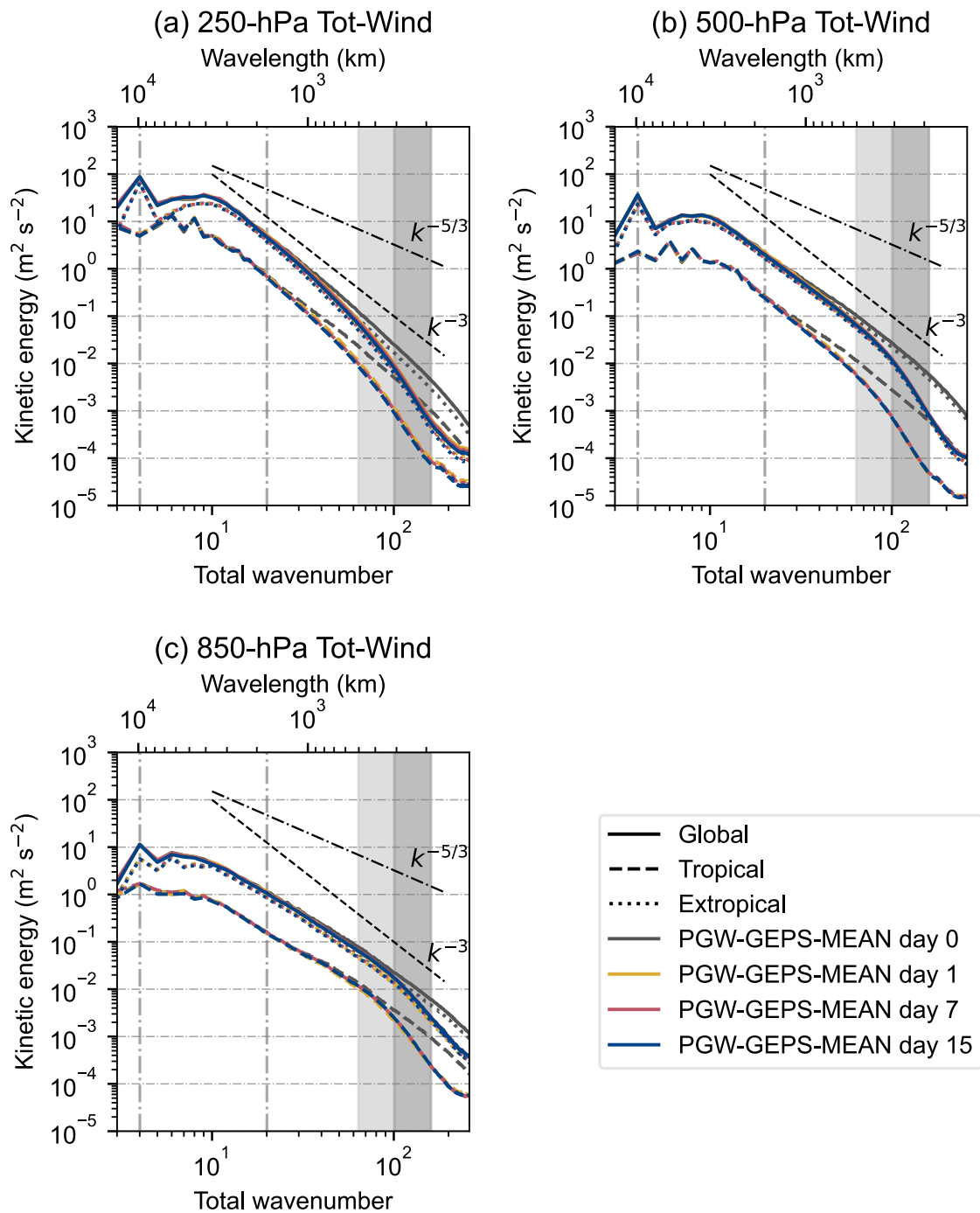


Fig. 5. KE spectra of all ensemble members of PGW-GEPS at different forecast lead times, in which the solid line represents the global region, the dashed line the tropical region, and the dotted line the extratropical region. The spectra are displayed across the isobaric surfaces of (a) 250 hPa, (b) 500 hPa, and (c) 850 hPa. The sub-synoptic-scale (total wave number: $64 \leq n \leq 100$) and mesoscale (total wave number: $101 \leq n \leq 160$) ranges are indicated. Slope lines of $-5/3$ and -3 are marked in each subplot for reference, calculated from the average of all cases.

GEPS below the sub-synoptic scale. In other words, the SV IC perturbations cannot change the main structures of PGW-GEPS-CTL. This reduction behavior of the KE spectra in PGW-GEPS is not sensitive to the SV IC perturbations but is sensitive to the PGW models for inferencing.

After briefly discussing the results above, the KE spectra of CMA-GEPS at the forecast lead time are similar to those of CMA-GEPS-CTL and ERA5 at the initial time. The SV IC and model perturbations can barely influence the KE spectra of CMA-GEPS compared to CMA-GEPS-CTL. In contrast to PGW-GEPS, there is a noticeable KE reduction below the sub-synoptic scale when the 24-h PGW forecasting model integrates. However, the KE at the sub-synoptic scale accounts for only 0.1% of the total KE, which means the main structures of global ensemble forecasts generated from PGW-GEPS are comparable to CMA-GEPS (see section 4.2.1). This is because the KE spectra of PGW-GEPS are nearly the same as those of CMA-GEPS and ERA5 beyond the sub-synoptic scale. In addition, the reasons for the reduction behavior cannot be attributed to ERA5 (the training data) and ICs for inferencing, due to the fact that ERA5 and CMA-GEPS do not show a noticeable reduction. In summary, the behavior of the KE reduction means that the effective resolution of PGW-GEPS is beyond the sub-synoptic scale.

Comparing the KE spectra of the ensemble mean of CMA-GEPS at different forecast lead times (Fig. S7 in the ESM), it can be seen that the KE spectra are attenuated as the lead time increases. However, the KE of the ensemble mean is declining fast at the mesoscale during the short-range forecast lead time. As the lead time increases, the larger-scale KE declines faster during the medium-range forecast lead time. Ultimately (beyond seven days), only the KE at a large scale shows a significant decrease. Thus, the KE spectra of the ensemble mean of CMA-GEPS show “flow-dependent” behavior. However, the reduction behavior of PGW-GEPS only appears under the sub-synoptic scale with the lead time increasing.

Figure 6 illustrates that the reduction behavior of PGW-GEPS can also influence the KE spectra of the ensemble mean. Specifically, the noticeable decrease occurs in the short-range forecast lead time. Meanwhile, the KE spectra of the ensemble mean are similar to those of CMA-GEPS beyond the sub-synoptic scale during the medium-range forecast lead time.

The temporal evolution of SKE spectra of CMA-GEPS and PGW-GEPS are shown in Fig. 7 and Fig. 8, respectively. It can be seen that the SKE of CMA-GEPS and PGW-GEPS have similar growth behavior with comparable amplitude. Specifically, the SKE grows faster under the synoptic scale during the short-range forecast lead time. Until the OTI (48 h), the SKE is saturated in the sub-synoptic scale. In addition, at the synoptic scale, the SKE saturates at the lead time of five days. After that, the peaks of SKE decrease from a higher wavenumber to a lower wavenumber with the lead time increasing, which means the SKE spectra grow “up-scale.” Meanwhile, the SKE also grows “up-amplitude”

and uniformly on a large scale. In conclusion, the SKE growth behaviors of PGW-GEPS and CMA-GEPS are similar, especially beyond the sub-synoptic scale. However, the SKE of PGW-GEPS is lower than the background spectra below the mesoscale. Even the background spectra of PGW-GEPS are lower than those of CMA-GEPS at the mesoscale.

The rotational wind structures of SKE are identical to those of total wind (figures not shown for brevity). This is because the growth of the SV IC perturbation is associated with the development of baroclinic weather systems, and the rotational wind roughly approximates balanced flow.

The EKE spectra of CMA-GEPS (Fig. 9) and PGW-GEPS (Fig. 10) are compared at different forecast lead times (using CMA-GEPS-CTL as the reference truth). It can be seen that the EKE spectra grow “up-scale” with “up-amplitude” at the large scale. Additionally, the EKEs of CMA-GEPS are saturated in the short-range forecast lead time. Due to the SV IC perturbations growing with the OTI, the peak of EKE is close to SKE at 48 h. Subsequently, the EKE’s and SKE’s amplitudes and growth behaviors are nearly identical at the medium-range forecast lead time. It should be noted that the EKE growth behaviors of PGW-GEPS are similar to those of CMA-GEPS beyond the sub-synoptic scale. However, the EKE of PGW-GEPS (Fig. 10) is significantly larger than the background spectra under the sub-synoptic scale. These characteristics are inconsistent with CMA-GEPS. According to the results above, PGW-GEPS cannot predict mesoscale motions that lead to inconsistency between SKE and background spectra.

The following is a brief discussion and a summary of the conclusions that can be drawn from this section:

- (1) The ensemble forecasts of PGW-GEPS are sensitive to the SV IC perturbations at the medium range.
- (2) The SV IC perturbations can grow realistically in PGW-GEPS beyond the sub-synoptic scale. This is because PGW-GEPS can predict synoptic-scale and large-scale atmospheric motion reasonably compared to CMA-GEPS.
- (3) The KE reduction of PGW-GEPS under the sub-synoptic scale makes the ensemble forecasts smoother than in CMA-GEPS. Meanwhile, the ensemble mean, SKE, and EKE growth behaviors of PGW-GEPS are inconsistent with those of CMA-GEPS below the sub-synoptic scale.
- (4) The interpolation cannot be the reason for this reduction because Selz and Craig (2023) and Bonavita (2024) showed similar characteristics of the KE spectra of PGW calculated from 0.25° grid data (the same resolution as PGW). Additionally, the KE spectra of ERA5 training data and ICs are similar without decreasing at the sub-synoptic scale. The minimized L1 or L2 loss function in training the DL-based model may lead to the DL-based model paying attention to the global weather forecast skill beyond the sub-synoptic scale. This is because the growth from the initial uncertainties at large scales appears dominant over the impact of errors cascading up from small scales (Žagar, 2017).
- (5) The reduction behavior also occurs in some global

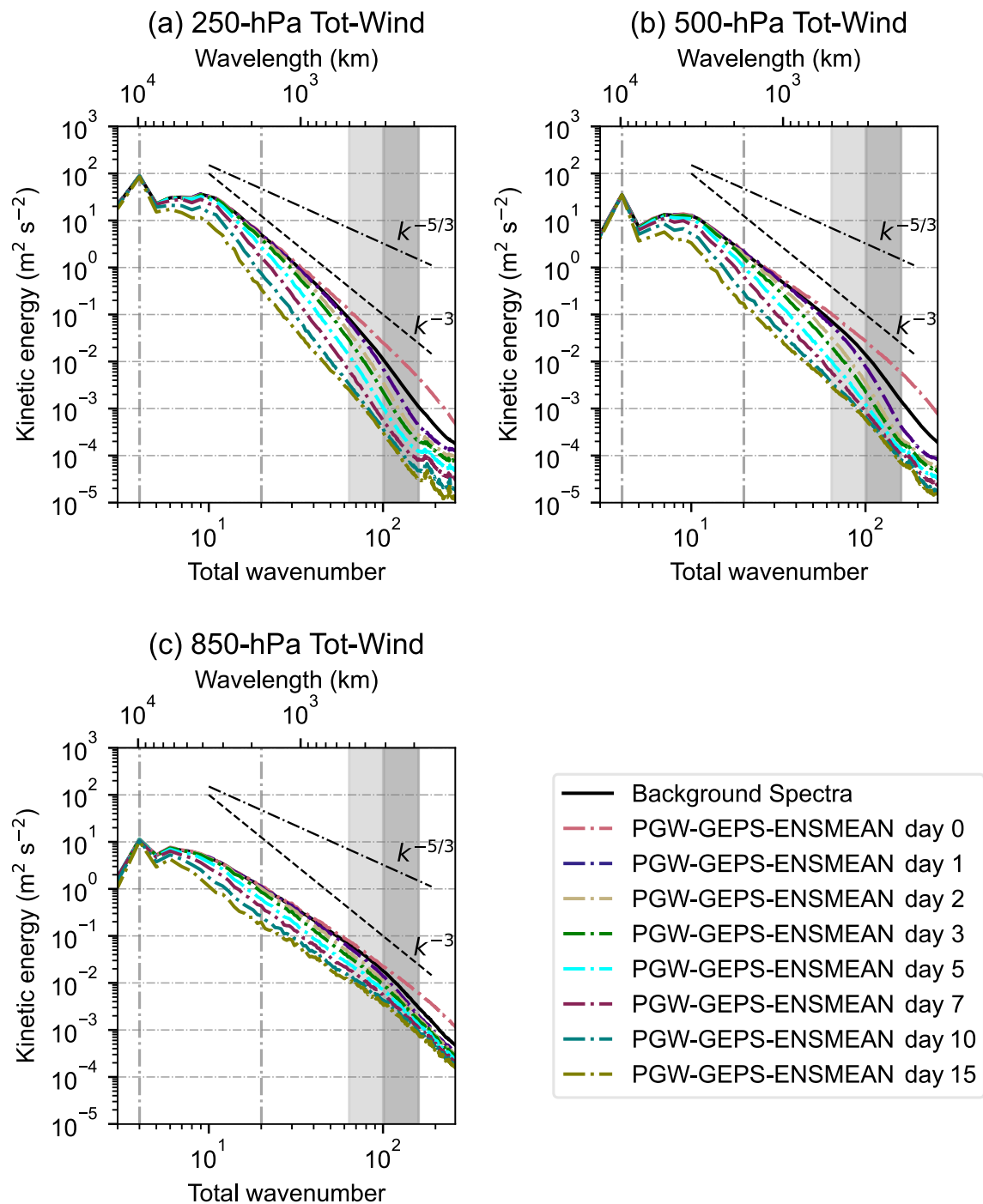


Fig. 6. KE spectra of the ensemble mean of PGW-GEPS at different forecast lead times. Black solid lines represent the background spectra, which are calculated from the KE of CMA-GEPS averaged across all forecast lead times with all ensemble members, and dashed lines represent the KE of the ensemble mean at the forecast lead time. The spectra are displayed across the isobaric surfaces of (a) 250 hPa, (b) 500 hPa, and (c) 850 hPa. The sub-synoptic scale (total wave number: $64 \leq n \leq 100$) and mesoscale (total wave number: $101 \leq n \leq 160$) ranges are indicated. Slope lines of $-5/3$ and -3 are marked in each subplot for reference, calculated from the average of all cases.

NWP models with similar horizontal spacing (Wang and Sardeshmukh, 2021). Thus, this behavior only indicates that the effective resolution of PGW-GEPS is beyond the sub-synoptic scale, limited to predicting mesoscale atmospheric motions. However, many extreme events are associated

with sub-synoptic-scale weather systems.

(6) In addition, PGW-GEPS can generate a realistic ensemble spread based on SV IC perturbations because PGW can provide comparable skillful forecasts compared to traditional NWP beyond the sub-synoptic scale.

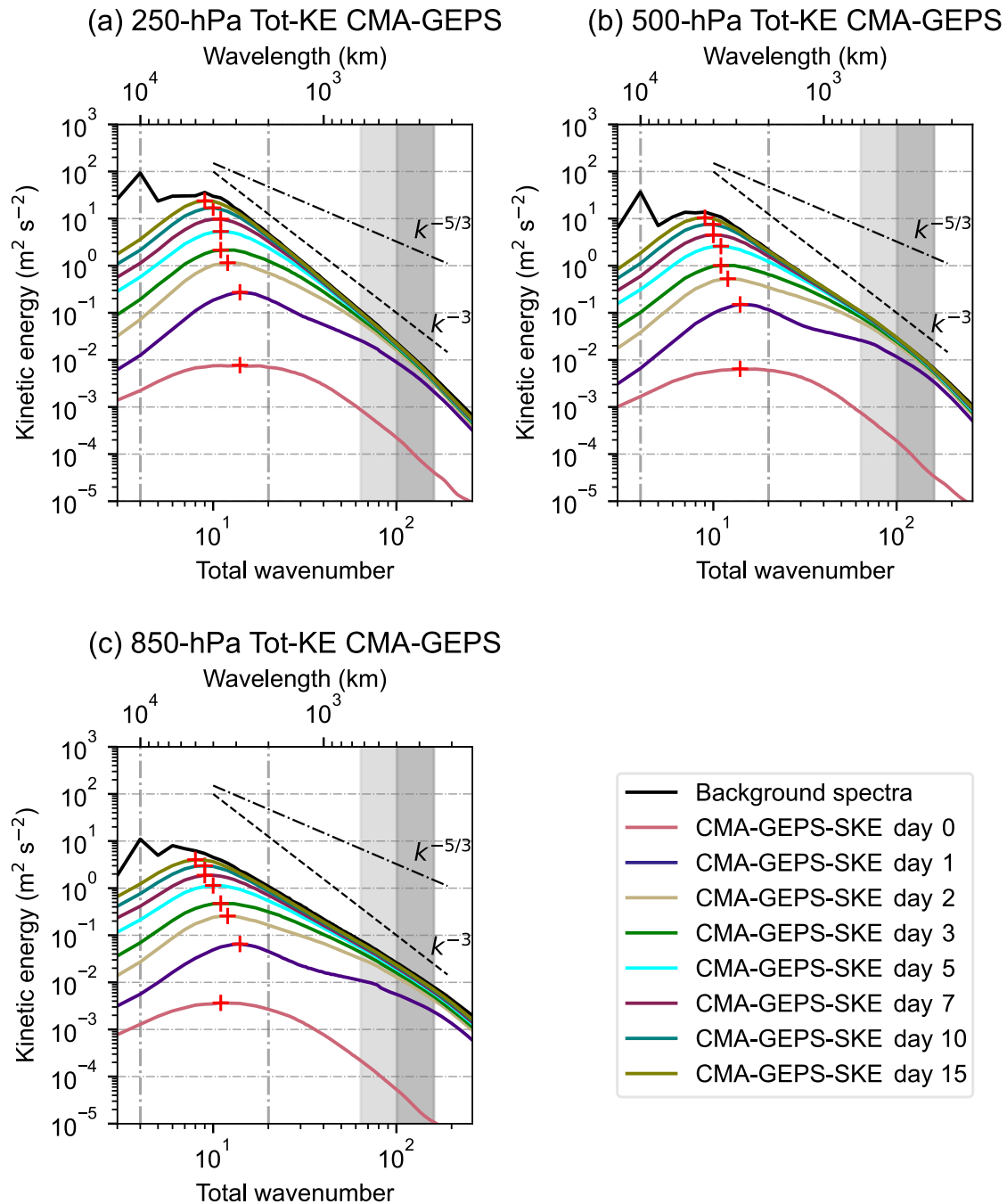


Fig. 7. SKE spectra of CMA-GEPS at different forecast lead times. Black solid lines represent the background spectra, calculated from the KE of CMA-GEPS averaged across all forecast lead times with all ensemble members (highlighted for comparison), and other lines represent the SKE at the forecast lead time. The spectra are displayed across the isobaric surfaces of (a) 250 hPa, (b) 500 hPa, and (c) 850 hPa. The sub-synoptic-scale (total wave number: $64 \leq n \leq 100$) and mesoscale (total wave number: $101 \leq n \leq 160$) ranges are indicated. Slope lines of $-5/3$ and -3 are marked in each subplot for reference, calculated from the average of all cases.

4.3. Global ensemble prediction performance of PGW-GEPS and CMA-GEPS

In this section, we investigate the global medium-range ensemble prediction performance of PGW-GEPS and CMA-GEPS. For fair verification, both CMA-GEPS and PGW-GEPS forecasts are verified against the CMA-4DVAR opera-

tional analysis at 1.5° grid resolution following the WMO guidelines (Mylne et al., 2022). All the forecast metrics are calculated from the average of all cases (total 40), and a bootstrap and Student's t -test algorithm (Hamill, 1999; Zhu et al., 2023) are used to test the significance of the score differences between CMA-GEPS and PGW-GEPS.

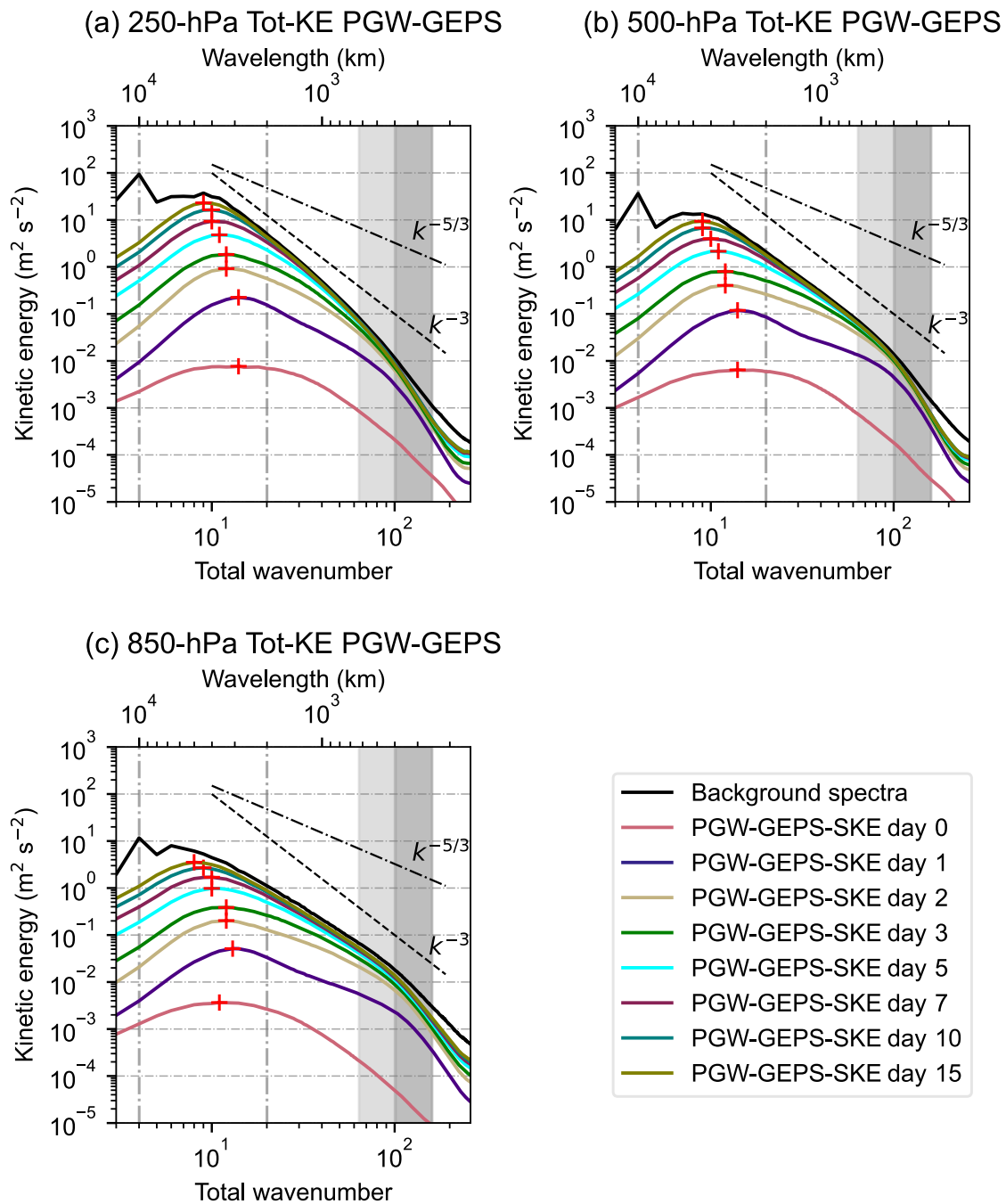


Fig. 8. As in Fig. 7 but for SKE spectra of PGW-GEPS. Note the background spectra are calculated from PGW-GEPS.

It can be seen from Fig. 11 that PGW-GEPS can provide comparable global medium-range ensemble forecasts to CMA-GEPS according to deterministic and probabilistic verification metrics. Additionally, PGW-GEPS performs better in medium-range forecasting in Trpc than CMA-GEPS.

In order to better understand the effect of SV IC perturbations on PGW-GEPS, the following analysis focuses on the details of RMSE and spread at all lead times. Figure 12 shows the averaged RMSE and spread for different variables over NH and SH. It can be seen from Fig. 12 that RMSEs

increase steadily with lead time, and the RMSE of the ensemble mean forecasts from all experiments is lower than that of CTL during the medium-range forecast period. Therefore, applying SV IC perturbations in PGW-GEPS can significantly reduce the RMSE during medium-range forecasting compared to PGW deterministic forecasting. When compared to CMA-GEPS, the RMSEs of PGW-GEPS are higher during the short range, especially at the first lead time. This is probably due to the inevitable uncertainty between CMA-4DVAR and ERA5. In addition, the RMSEs of PGW-GEPS are

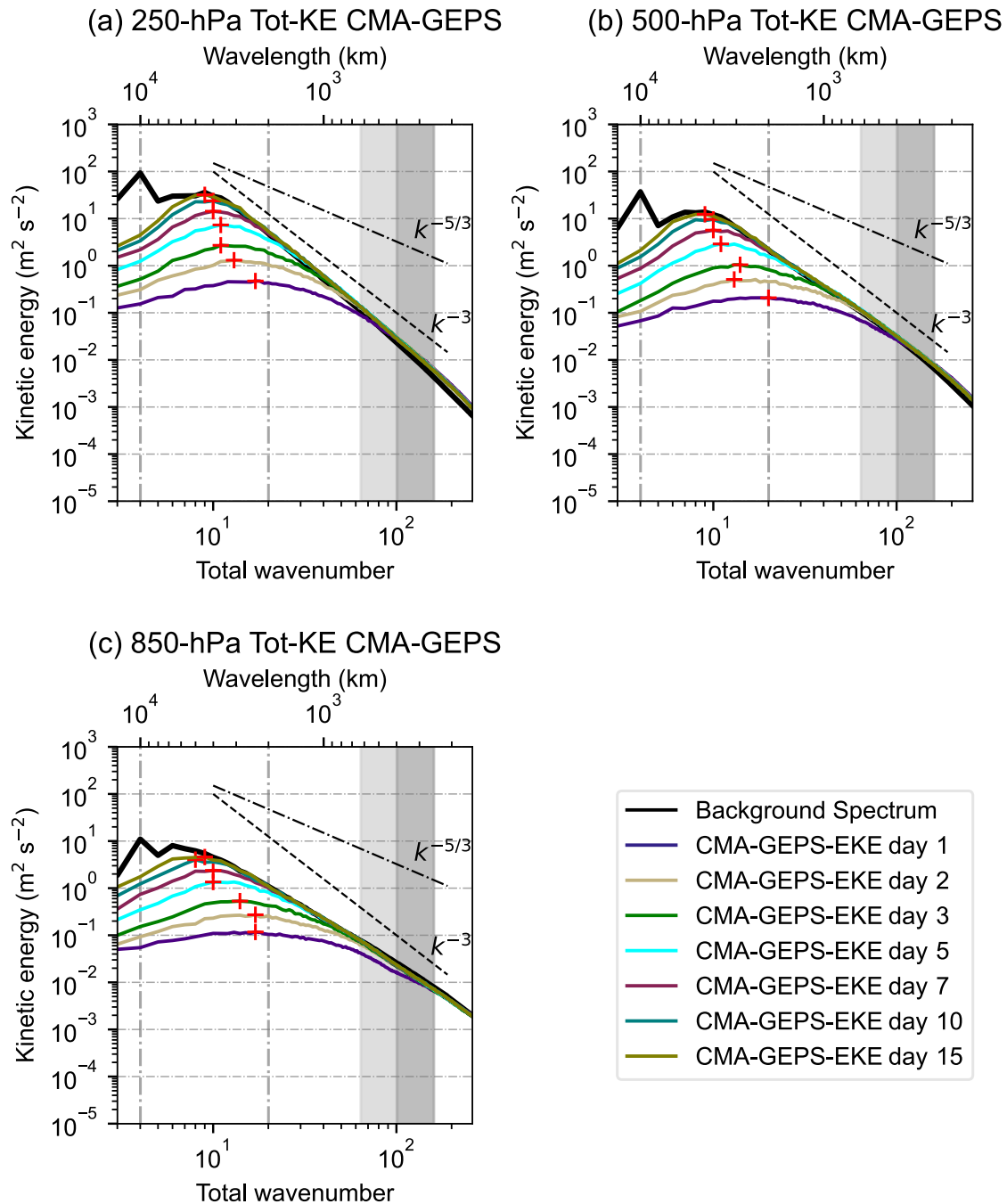


Fig. 9. As in Fig. 7 but for EKE spectra of CMA-GEPS.

lower than CMA-GEPS during the medium-range forecasting for upper variables (250 and 500 hPa). However, the differences are not statistically significant (95% confidence level). For the variables at 500 hPa in NH, the RMSEs of PGW-GEPS are slightly (75% confidence level) better than those of CMA-GEPS during day 6 to day 10. For the variables at 250 hPa and 500 hPa in SH, the RMSEs of PGW-GEPS are slightly (75% confidence level) better than those of CMA-GEPS during day 10 to day 14. For the variables at 850 hPa in NH, the RMSEs of CMA-GEPS are significantly better than PGW-GEPS at the lead time and slightly better

during the short range in SH. In addition, the RMSE of PGW-RP-GEPS is lower than that of CMA-GEPS-CTL and the of PGW-GEPS-CTL during the medium-range forecast lead time. However, the RMSE of PGW-RP-GEPS is significantly higher than that of CMA-GEPS and PGW-GEPS. Similar conclusions can be drawn based on radiosonde observations [see Fig. S8 (in the ESM) for details].

The ensemble spread expresses the distance between the ensemble members and the ensemble mean to indicate the range and amplitude of the perturbation. Therefore, the spread should be large enough and should also maintain a rea-

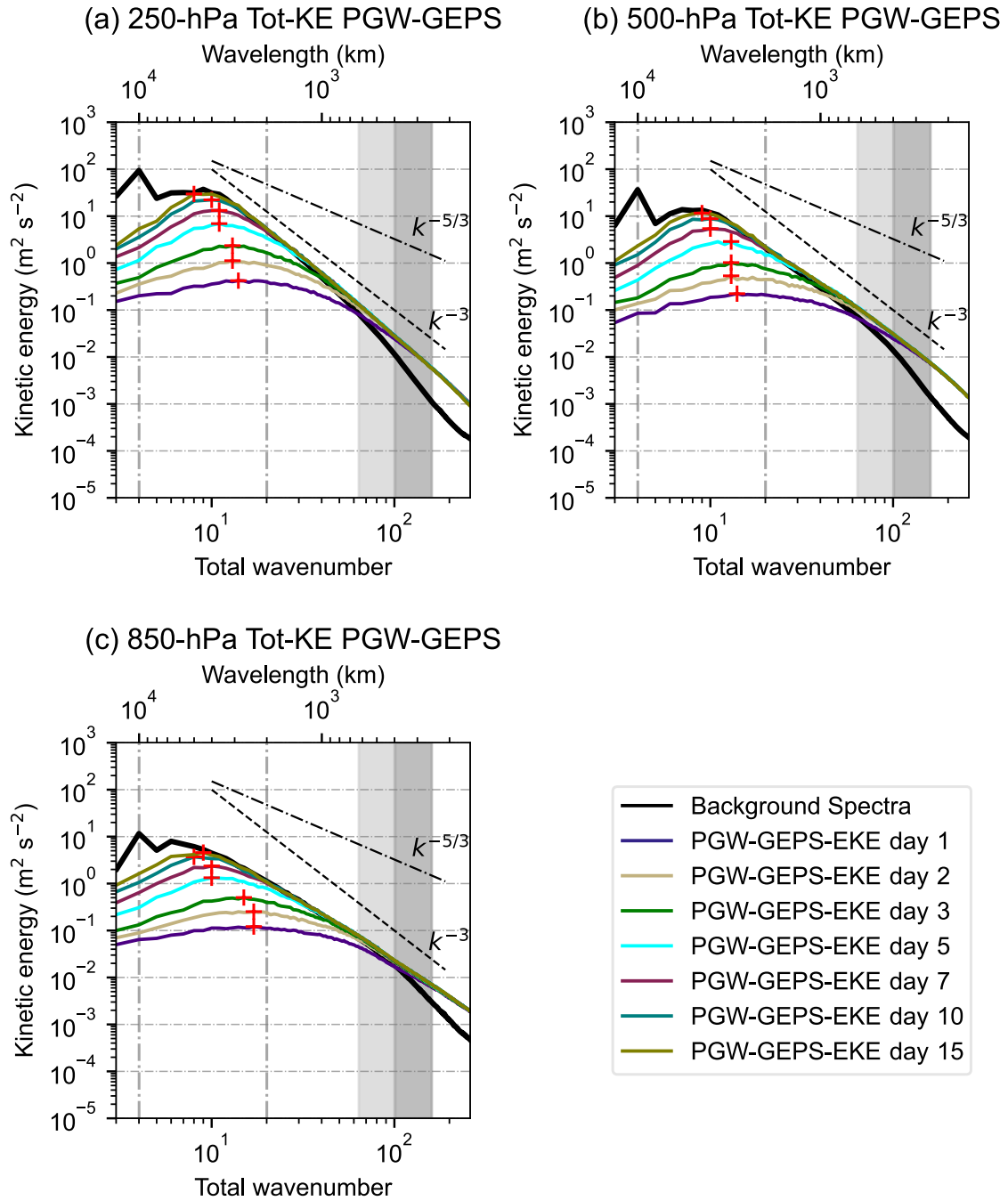


Fig. 10. As in Fig. 9 but for EKE spectra of PGW-GEPS. Note the background spectra are calculated from PGW-GEPS.

sonable proportionality with the RMSE to ensure the predictability of the GEPS. It can be seen from Fig. 12 that all experiments' ensemble spread continues to increase with the lead time. The ensemble spread of CMA-GEPS and PGW-GEPS is nearly the same in NH. However, the ensemble spread of PGW-GEPS is lower than that of CMA-GEPS. Notably, the ensemble spread of PGW-RP-GEPS grows more slowly compared to both CMA-GEPS and PGW-GEPS, which affects the balance between spread and RMSE. Ideally, a well-calibrated ensemble system should

maintain a proportional relationship between these two metrics, but in the case of PGW-RP-GEPS, this balance is weaker, potentially limiting its predictive ability. Overall, their ensemble spreads are comparable to each other. This slower growth in spread suggests that the random perturbations applied in PGW-RP-GEPS may not evolve as dynamically as those generated through more sophisticated techniques, like SV IC perturbations. In conclusion, the SV IC perturbation in PGW-GEPS can grow with baroclinic system development, which leads to the spread of outcomes growing

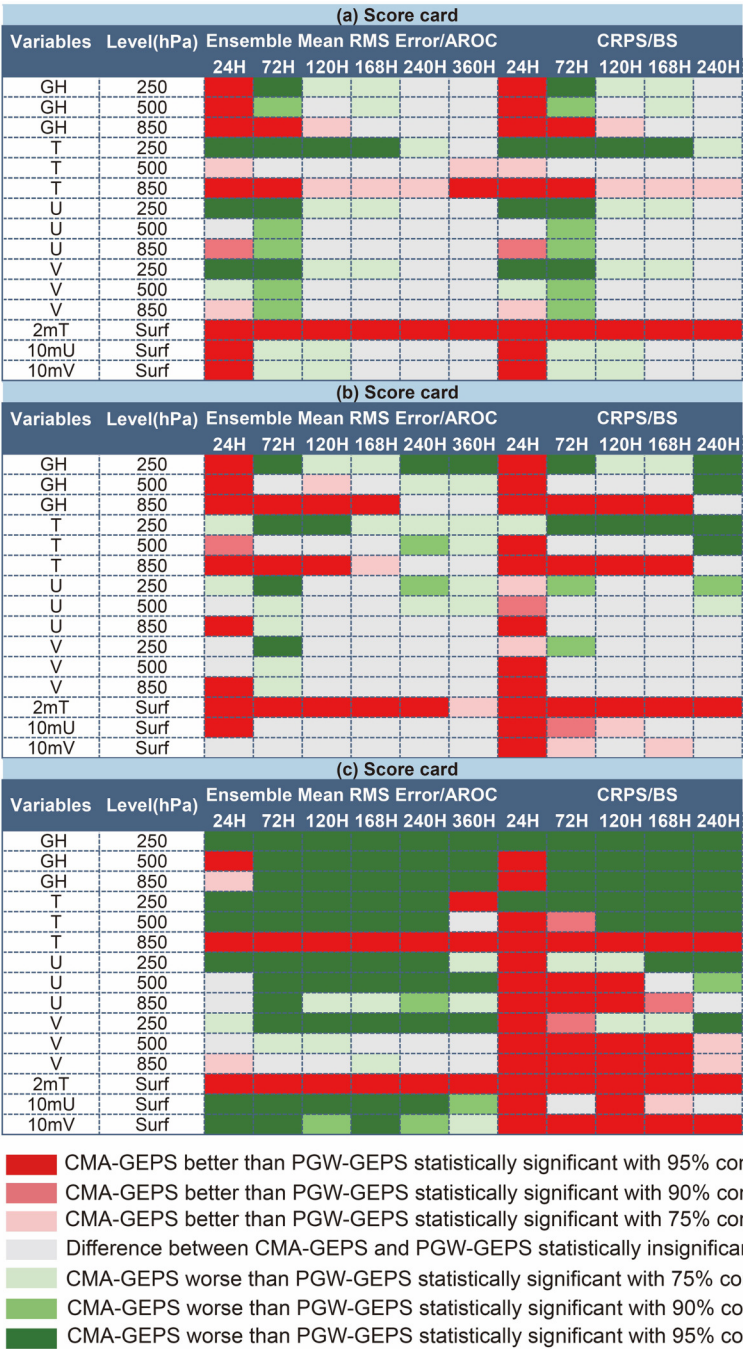


Fig. 11. Scorecard comparing CMA-GEPS and PGW-GEPS in (a) NH, (b) SH, and (c) Trpc, verified against the CMA-4DVAR analysis at the forecast lead time. Red/green indicates CMA-GEPS is better/worse than PGW-GEPS at the confidence interval (the significance interval is tested by Student’s *t*-test algorithm). Gray means no significant difference. The results are calculated from the average of all cases.

realistically with sufficient dispersion.

The continuous ranked probability score (CRPS) is calculated by the sum of the squared difference between the cumulative distribution function (CDF) of the forecasted probabilities and the CDF of the observations. Figure 13 shows the CRPSs of CMA-4DVAR and PGW-GEPS, from which it can be concluded that the CRPSs of CMA-GEPS are lower than those of PGW-GEPS during the medium-range lead

time. Specifically, PGW-GEPS is slightly better than CMA-GEPS during the middle or late forecast period in the medium range for upper-level variables at 250 hPa and 500 hPa, especially in SH; but significantly worse for upper-level variables at 850 hPa. Due to the poorer RMSE and spread performance of PGW-RP-GEPS compared to CMA-GEPS and PGW-GEPS, it is not included in the CRPS evaluation for clarity. Therefore, the application of SV IC perturba-

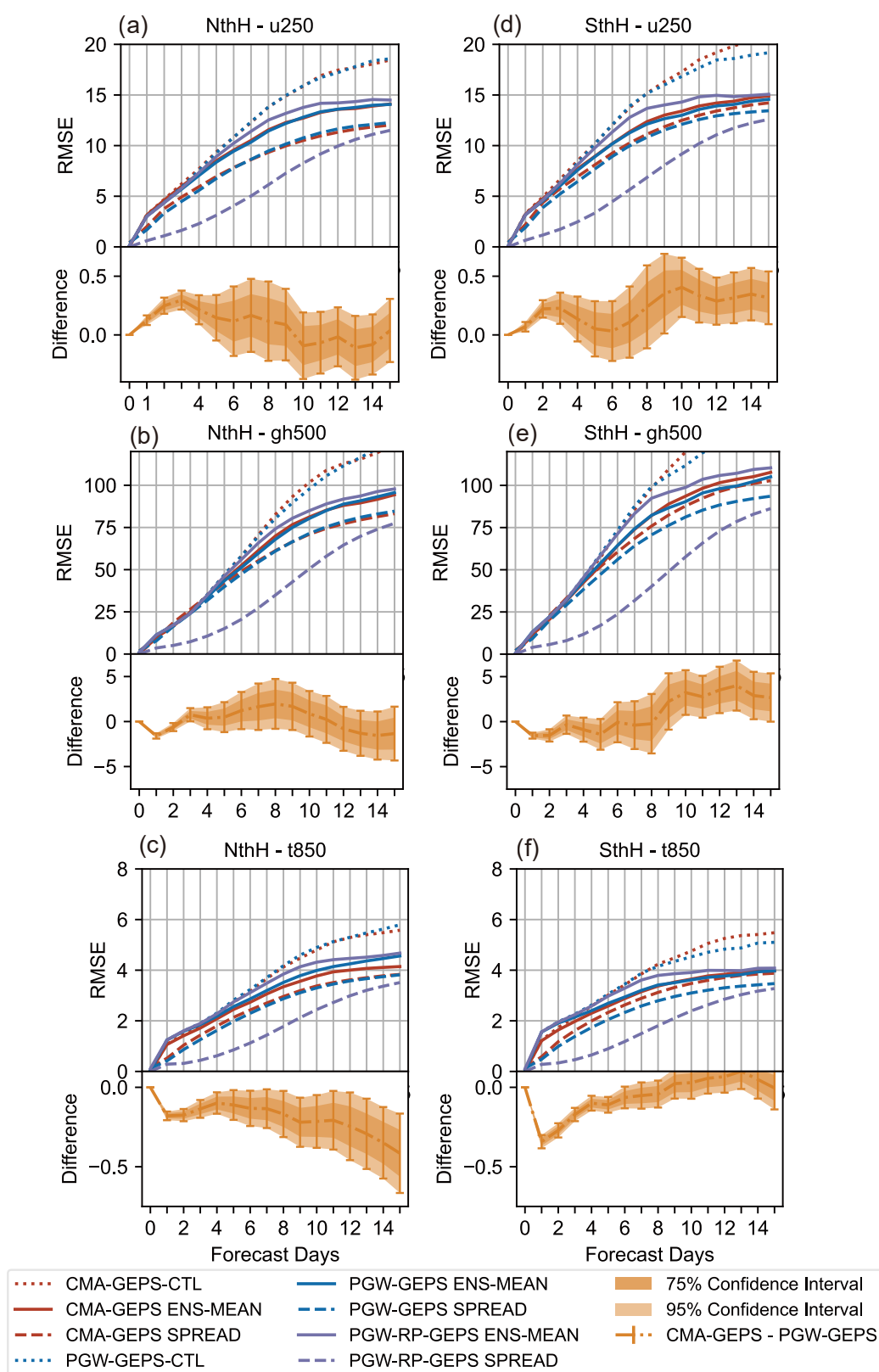


Fig. 12. Time series of RMSEs (solid line) and the ensemble spread (dashed line) of CMA-GEPS (red lines), PGW-GEPS (blue lines), and PGW-RP-GEPS (purple lines) for the (a, d) 250-hPa u wind speed, (b, e) 500-hPa geopotential height, and (c, f) 850-hPa temperature over (a–c) NH (20° – 80° N) and (d–f) SH (20° – 80° S). The RMSE differences (CMA-GEPS minus PGW-GEPS) are significant at the 75% and 95% confidence level when the values are outside the reference line (light-gray dashed line). The results are calculated from the average of all cases.

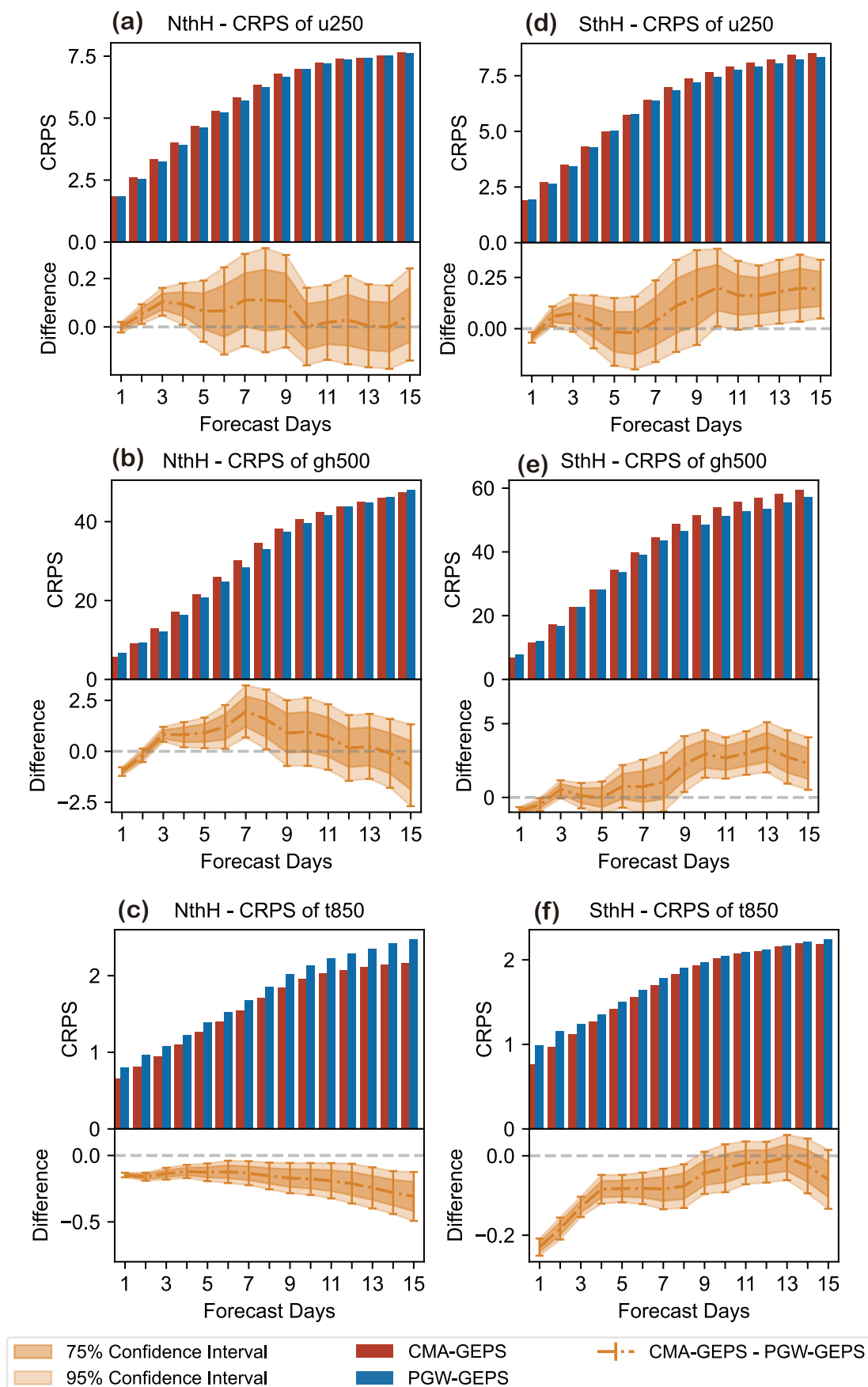


Fig. 13. As in Fig. 12 but for CRPS verifications.

tions in PGW-GEPS makes the skill of probability forecasts comparable to CMA-GEPS in the upper layers.

To further analyze the prediction skill of CMA-GEPS and PGW-GEPS, Figure 14 presents the ACC of the 500-hPa geopotential height over NH and SH. When the ACC is closer to 1, it means a higher level of prediction skill. It can be seen from Fig. 14 that the results of the ensemble experiments are better than the CTL, indicating that PGW-GEPS's forecast accuracy and skill significantly improve compared to PGW deterministic forecasts ($ACC > 0.6$, increase of 1.4 days). In addition, the ACCs of PGW-GEPS are higher than for CMA-GEPS during the medium range, but all the differences are not statistically significant. Specifically, when $ACC = 0.6$, PGW-GEPS can provide valuable ensemble forecasts in NH at a lead time of 9.4 days, higher than the 9.1 days for CMA-GEPS; and at lead time of 8.9 days in SH, higher than the 8.7 days for CMA-GEPS. Thus, the prediction skill of PGW-GEPS is comparable to that of CMA-GEPS. Moreover, the usable forecast days of PGW-RP-GEPS in NH and SH are 8.5 days and 8 days, respectively. This indicates that the forecasting skill of PGW-RP-GEPS is better than the control forecast but lower than both CMA-GEPS and PGW-GEPS.

The following is a brief discussion and a summary of the conclusions that can be drawn from this section:

(1) The global medium-range ensemble prediction skill of PGW-GEPS is comparable to that of CMA-GEPS in the extratropics.

(2) The main reason PGW-GEPS performs better than

CMA-GEPS in the tropics is that the bias of PGW-GEPS is lower than that of CMA-GEPS (figures not shown for brevity).

(3) The results of PGW-GEPS also indicate that the DL-based model is sensitive to the ICs. Using the training data for inferencing could improve performance. However, PGW has a generalization ability to provide skillful global medium-range forecasts with different ICs from NWP (model-dependent).

(4) In addition, the verification score is sensitive to the reference truth (analysis or reanalysis) in the short-range forecast lead time, especially in the lower layers and Trpc (Park et al., 2008). Thus, if ERA5 is used as the reference truth (figures not shown for brevity), the verification score of PGW-GEPS will improve in the short range, especially in the lower layers and Trpc.

5. Conclusions and discussion

The main goals of this paper were (1) to investigate the ensemble forecast sensitivity of DL-based models to IC errors, and (2) to recognize the difference between the DL-based model and operational GEPS forecasts when using the same IC perturbations. In order to achieve these goals, we used the SV IC perturbations for a DL-based model, PGW, in global medium-range weather ensemble prediction (PGW-GEPS). We carried out the ensemble prediction experiments in different seasons. Meanwhile, CMA-GEPS forecasts were used as the benchmark for comparison. The main

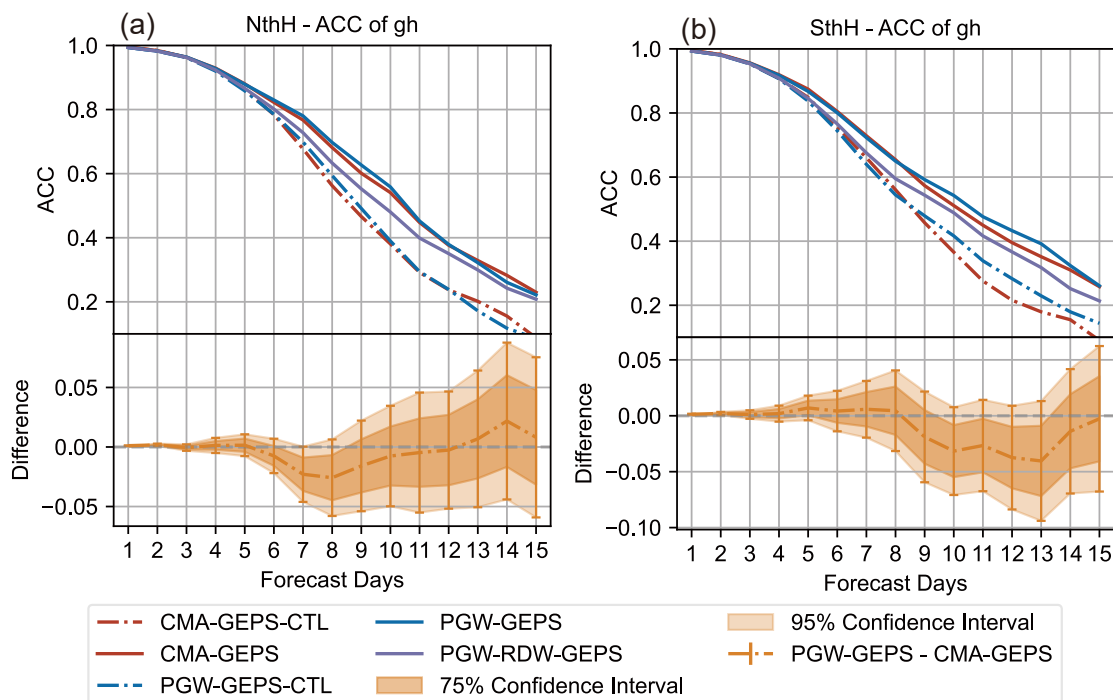


Fig. 14. Time series of ACC scores of CMA-GEPS (red), PGW-GEPS (blue), and PGW-RP-GEPS for the 500-hPa geopotential height over (a) NH (20°–80°N) and (b) SH (20°–80°S). The ACC differences (CMA-GEPS minus PGW-GEPS) are significant at the 95% (slightly at the 75%) confidence level when the values are outside the reference line (light-gray dashed line). The results are calculated from the ensemble mean averaged for all cases.

results and conclusions can be summarized as follows:

(1) The ensemble forecasts of PGW-GEPS are sensitive to the SV IC perturbations at the medium range. Meanwhile, the SV IC perturbations can grow realistically in PGW-GEPS beyond the sub-synoptic scale. This is because PGW-GEPS can predict synoptic-scale and large-scale atmospheric motion reasonably compared to NWP.

(2) However, the KE reduction of PGW-GEPS under the sub-synoptic scale makes the ensemble forecasts smoother than in CMA-GEPS. Meanwhile, the ensemble mean, SKE, and EKE growth behaviors of PGW-GEPS are inconsistent with those of CMA-GEPS below the sub-synoptic scale. However, the reduction behavior also occurs in some global NWP models with similar horizontal spacing. Nevertheless, this behavior indicates that the effective resolution of PGW-GEPS is beyond the sub-synoptic scale and is limited to predicting mesoscale atmospheric motions.

(3) The global medium-range ensemble prediction skill of PGW-GEPS is comparable to that of CMA-GEPS in the extratropics. The results of PGW-GEPS also indicate that PGW has a general ability to provide skillful global medium-range forecasts with different ICs from NWP.

These results highlight the potential for combining traditional NWP IC perturbation approaches with emerging DL-based models for global medium-range weather prediction. Comparable forecast skill was achieved with greatly improved computational efficiency, requiring just one GPU for inference in PGW-GEPS, and its computing time was half that of CMA-GEPS. In addition, the DL-based model could provide comparable global medium-range probabilistic forecasts for NWP.

It is also important to emphasize that DL-based models possess several advantages, particularly in their backpropagation modules. Due to DL-based models being able to apply the chain rule to known linear and nonlinear functions, they are well suited for automatic differentiation (Vonich and Hakim, 2024). The backpropagation modules allow DL models to be promising candidates for directly calculating sensitivity vectors or other types of perturbations. However, there are several challenges that need to be addressed when applying DL models to generate perturbations. One of the primary concerns is the physical consistency of DL models. In contrast to traditional models, where physical laws explicitly govern the dynamics, DL models may not fully capture or adhere to the underlying physical principles, potentially resulting in inconsistencies. This issue has been highlighted not only by our findings but also by previous studies, such as those of Selz and Criag (2023) and Bonavita (2024), who pointed out that current DL-based models still face limitations in maintaining physical consistency. Thus, when using DL models for adjoint calculations, we suggest considering the correlations and consistency between forecast variables and the reasonableness of perturbations.

Another challenge for global DL-based weather forecast models is that their effective resolution is often limited to mesoscale atmospheric motions, particularly beyond the

sub-synoptic scale. In other words, DL-based models need to pay more “attention” to capturing sub-synoptic scale motions during training to improve further. Notably, generative DL methods have become increasingly popular for developing ensemble DL-based weather prediction models (Price et al., 2024; Li et al., 2024a; Zhong et al., 2024). An auspicious example is the ECMWF’s ensemble AIFS (Artificial Intelligence/Integrated Forecasting System), which, surprisingly, does not suffer from the excessive smoothing typically seen in deterministic DL-based models (Lang et al., 2024). This generative approach has shown significant potential in improving the DL-based model by maintaining the variability of atmospheric processes at different scales.

Furthermore, the uncertainty of ICs used as the input for DL-based models is inevitable. This can be addressed by providing IC perturbations from NWP models. It is equally important to account for uncertainties in training and real-time input data used to train and drive these models. This is because the models used to generate reanalysis data often differ from those used in operational forecasting. This can introduce systematic biases that must be accounted for when training DL models for medium-range forecasts.

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